Chapter S

Additional files

This document holds the tables and figures that support the main chapters but are too long to include in them. They are grouped by the chapter they support and numbered in order as Additional files.

S.1 Noise robustness across molecular representations and model architectures

Additional file 1: Hyperparameters for every model configuration, and the descriptors that make up PDV

Table A lists the default settings, which both the QM9 and the assay pipelines read from one file. Table B lists the tuned settings that replace a default, per model and per dataset, on QM9, LogD, Caco-2 and hERG K_i. Table C lists the 200 descriptor names that make up PDV, in the order they are computed.

Table S.1: Table A: default hyperparameters for every model configuration in the study. Every value is read from `models/model_defaults.py` at spec version 1.7.0, spec hash `9249926c38dc`, which is the file both the QM9 pipeline and the assay pipeline load their settings from. Every QM9 results row carries the spec version and the spec hash as two of its columns, so a QM9 table can be traced back to the settings that produced it; the three assay datasets are written by a second pipeline whose results files do not carry those two columns. The configuration list is the model roster the job generator submits. One setting is used for every representation and for every dataset, except where Table B gives a tuned replacement. The support vector machine uses an RBF kernel on every representation, so no result in this study carries a kernel that changes with the representation. Two values are not held in the shared spec and name their source file in the third column.

Configuration	Hyperparameter	Default value
Shared by every neural network below		
Training, all networks	optimizer	adam
	lr	0.001
	weight_decay	0.0
	batch_size	32
	epochs	100 cap; early stopping usually reaches it first
	patience	10 epochs without a strictly better mean validation loss
	restore_best_weights	True the reported fit is the best epoch, not the last
	val_loss_reduction	mean
	improvement_tolerance	0.0
	mc_passes	100 stochastic forward passes at inference
	standardise_targets	True fitted on training labels alone
Tree and boosting models		
RF `rf`	n_estimators	100
	max_depth	None no limit
	min_samples_leaf	5
	min_samples_split	2
	max_features	0.3
	bootstrap	True
RF300 `rf300`	n_estimators	300
	max_depth	None no limit

Configuration	Hyperparameter	Default value
	min_samples_leaf	5
	min_samples_split	2
	max_features	0.3
	bootstrap	True
XGBoost `xgboost`	n_estimators	100
	max_depth	6
	learning_rate	0.1
	subsample	1.0
	colsample_bytree	1.0
	colsample_bylevel	1.0
	min_child_weight	1
	gamma	0.0
	reg_alpha	0.0
	reg_lambda	1.0
LightGBM `lgb`	n_estimators	100
	learning_rate	0.1
	num_leaves	31
	max_depth	-1 no limit
	subsample	1.0
	colsample_bytree	1.0
	min_child_samples	20
	reg_alpha	0.0
	reg_lambda	0.0
NGBoost `ngboost`	n_estimators	500
	learning_rate	0.01
	natural_gradient	True
	dist	Normal ngboost.distns.Normal
	score	MLE ngboost.scores.MLE, the log scoring rule
	minibatch_frac	1.0
	col_sample	1.0
	base_max_depth	3 depth of one boosting stage
	base_criterion	friedman_mse
	early_stopping_rounds	50 stages without a better validation log likelihood
	use_best_iteration	True predictions use the best stage, not every stage fitted

Configuration	Hyperparameter	Default value
	ngboost_ensemble_seeds	1 1 = one fit, no seed ensemble
QRF `qrf`	n_estimators	300
	max_depth	None no limit
	min_samples_leaf	5
	min_samples_split	2
	max_features	0.3
	bootstrap	True
Kernel models		
SVM `svm`	C	1.0
	gamma	scale
	kernel	rbf
GP `gauche_rbf`	kernel	rbf set by this job, not by the shared spec
	likelihood_noise	0.001 starting value; it is learned
	outputscale	1.0
	apply_outputscale	True
	max_train_n	5000 molecules the process is fitted on; an exact Gaussian process is cubic in this. The test set is untouched and the number actually fitted is written into every results row
	fallback_adam_lr	0.1 used only when the standard fitter refuses the model
	fallback_adam_iters	100 used only when the standard fitter refuses the model
	single_thread_fit	False a threading workaround, off since the environment check found one runtime

Configuration	Hyperparameter	Default value
	init_lengthscale_from_data	True the RBF length scale starts at the median distance between training molecules, not at the library default
	lengthscale_probe_n	500 molecules sampled to estimate that median
	collapse_fraction	0.05 a fit whose predictions vary by less than this fraction of the training label spread is recorded as collapsed rather than scored
GP-Tanimoto `gauche` run on ecfp4 only	kernel	tanimoto set by this job, not by the shared spec
	likelihood_noise	0.001 starting value; it is learned
	outputscale	1.0
	apply_outputscale	True
	max_train_n	5000 molecules the process is fitted on; an exact Gaussian process is cubic in this. The test set is untouched and the number actually fitted is written into every results row
	fallback_adam_lr	0.1 used only when the standard fitter refuses the model
	fallback_adam_iters	100 used only when the standard fitter refuses the model
	single_thread_fit	False a threading workaround, off since the environment check found one runtime

Configuration	Hyperparameter	Default value
	init_lengthscale_from_data	True the RBF length scale starts at the median distance between training molecules, not at the library default
	lengthscale_probe_n	500 molecules sampled to estimate that median
	collapse_fraction	0.05 a fit whose predictions vary by less than this fraction of the training label spread is recorded as collapsed rather than scored
GP-Hetero `heteroscedastic_gp`	kernel	rbf set by this job, not by the shared spec
	likelihood_noise	0.001 starting value; it is learned
	outputscale	1.0
	apply_outputscale	True
	max_train_n	5000 molecules the process is fitted on; an exact Gaussian process is cubic in this. The test set is untouched and the number actually fitted is written into every results row
	fallback_adam_lr	0.1 used only when the standard fitter refuses the model
	fallback_adam_iters	100 used only when the standard fitter refuses the model
	single_thread_fit	False a threading workaround, off since the environment check found one runtime

Configuration	Hyperparameter	Default value
	init_lengthscale_from_data	True the RBF length scale starts at the median distance between training molecules, not at the library default
	lengthscale_probe_n	500 molecules sampled to estimate that median
	collapse_fraction	0.05 a fit whose predictions vary by less than this fraction of the training label spread is recorded as collapsed rather than scored
	noise network hidden sizes	[64, 64] models/models.py, not in the shared spec
	noise network output	Softplus, plus 10^{-4} models/models.py, not in the shared spec
	Adam learning rate, process	0.1 models/models.py, not in the shared spec
	Adam learning rate, noise network	0.001 models/models.py, not in the shared spec
	noise network epochs	100 the shared epoch cap
Neural networks		
NN-α `dnn`	hidden_sizes	[128, 64]
	dropout_rate	0.2
	activation	relu
NN-β `mlp`	hidden_size	128
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
BNN-α `dnn_bnn_full`	hidden_sizes	[128, 64]
	dropout_rate	0.2
	activation	relu
	bnn_prior_mu	0.0
	bnn_prior_sigma	0.1

Configuration	Hyperparameter	Default value
	bnn_kl_weight	elbo 1 / number of training molecules
BNN-β `mlp_bnn_full`	hidden_size	128
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
	bnn_prior_mu	0.0
	bnn_prior_sigma	0.1
	bnn_kl_weight	elbo 1 / number of training molecules
VBLL-α `dnn_bnn_full_variational`	hidden_sizes	[128, 64]
	dropout_rate	0.2
	activation	relu
	vbll_prior_mu	0.0
	vbll_prior_sigma	1.0
	vbll_init_log_sigma	-3.0
	vbll_init_log_noise_var	-2.0
VBLL-β `mlp_bnn_full_variational`	hidden_size	128
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
	vbll_prior_mu	0.0
	vbll_prior_sigma	1.0
	vbll_init_log_sigma	-3.0
	vbll_init_log_noise_var	-2.0
VBLL-α (noise head) `dnn_bnn_full_variational_hetero`	hidden_sizes	[128, 64]
	dropout_rate	0.2
	activation	relu
	vbll_prior_mu	0.0
	vbll_prior_sigma	1.0
	vbll_init_log_sigma	-3.0
	vbll_init_log_noise_var	-2.0
	observation noise	predicted per molecule a variational noise head on the same features, started at the single-number value above
VBLL-β (noise head) `mlp_bnn_full_variational_hetero`	hidden_size	128
	num_hidden_layers	2

Configuration	Hyperparameter	Default value
	dropout_rate	0.2
	activation	relu
	vbll_prior_mu	0.0
	vbll_prior_sigma	1.0
	vbll_init_log_sigma	-3.0
	vbll_init_log_noise_var	-2.0
	observation noise	predicted per molecule a variational noise head on the same features, started at the single-number value above
BNN-α (variance head) `dnn_bnn_full_mve`	hidden_sizes	[128, 64]
	dropout_rate	0.2
	activation	relu
	bnn_prior_mu	0.0
	bnn_prior_sigma	0.1
	bnn_kl_weight	elbo 1 / number of training molecules
	output head	value and log variance two outputs
	loss	Gaussian negative log likelihood
	log variance clamp	$[-10, 10]$ scripts/loss_functions.py, not in the shared spec
BNN-β (variance head) `mlp_bnn_full_mve`	hidden_size	128
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
	bnn_prior_mu	0.0
	bnn_prior_sigma	0.1
	bnn_kl_weight	elbo 1 / number of training molecules
	output head	value and log variance two outputs
	loss	Gaussian negative log likelihood
	log variance clamp	$[-10, 10]$ scripts/loss_functions.py, not in the shared spec

Table S.2: Table B: the tuned settings that replace a default, and the four datasets they apply to. A setting is chosen once per model per dataset, not per representation, and is then used for all six representations: a two-way comparison across model and representation cannot be read if each cell carries its own hyperparameters. Only the Bayesian and variational networks are tuned; every other configuration in Table A trains at the default. A model with no row for a dataset kept the default on that dataset. Learning rates are printed to three significant figures; the source files hold the full value. Source files: `results/master_tuned_hyperparameters.json` for QM9 and `results/master_tuned_hyperparameters_lab.json` for the three assay datasets.

Dataset	Configuration	Hyperparameter	Tuned value
QM9	NN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
QM9	BNN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
QM9	VBLL-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
QM9	NN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
QM9	BNN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
QM9	VBLL-β	dropout_rate	0.357
		hidden_size	64
		lr	0.00119
		num_hidden_layers	1
Caco-2	NN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
Caco-2	BNN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
Caco-2	VBLL-α	activation	relu
		hidden_size1	64
		hidden_size2	32
Caco-2	NN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
Caco-2	BNN-β	dropout_rate	0.401
		hidden_size	64

Dataset	Configuration	Hyperparameter	Tuned value
		lr	0.00587
		num_hidden_layers	3
hERG K_i	NN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
hERG K_i	BNN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
hERG K_i	VBLL-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
hERG K_i	NN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
hERG K_i	BNN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
LogD	NN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
LogD	BNN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
LogD	VBLL-α	activation	tanh
		hidden_size1	256
		hidden_size2	32
LogD	NN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
LogD	BNN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
LogD	VBLL-β	dropout_rate	0.079
		hidden_size	128
		lr	0.00011
		num_hidden_layers	2

Table S.3: Table C: the 200 descriptors that make up the PDV representation. One entry is one feature of the vector, in the order the pipeline computes them. The list is fixed in the code rather than taken from RDKit's current default set, which grows between releases, so the same 200 features are built on every dataset and in every run. They were computed with RDKit 2022.09.5. Descriptors that return a non-finite value on a molecule have that value replaced by zero before the standardization constants are computed. Source: `DEFAULT_DESCRIPTOR_LIST` in `scripts/process_and_train.py`.

BalabanJ	NOCount	SlogP_VSA7	fr_bicyclic
BertzCT	NumAliphaticCarbocycles	SlogP_VSA8	fr_diazo
Chi0	NumAliphaticHeterocycles	SlogP_VSA9	fr_dihydropyridine
Chi0n	NumAliphaticRings	TPSA	fr_epoxide
Chi0v	NumAromaticCarbocycles	VSA_EState1	fr_ester
Chi1	NumAromaticHeterocycles	VSA_EState10	fr_ether
Chi1n	NumAromaticRings	VSA_EState2	fr_furan
Chi1v	NumHAcceptors	VSA_EState3	fr_guanido
Chi2n	NumHDonors	VSA_EState4	fr_halogen
Chi2v	NumHeteroatoms	VSA_EState5	fr_hdrzine
Chi3n	NumRadicalElectrons	VSA_EState6	fr_hdrzone
Chi3v	NumRotatableBonds	VSA_EState7	fr_imidazole
Chi4n	NumSaturatedCarbocycles	VSA_EState8	fr_imide
Chi4v	NumSaturatedHeterocycles	VSA_EState9	fr_isocyan
EState_VSA1	NumSaturatedRings	fr_Al_COO	fr_isothiocyan
EState_VSA10	NumValenceElectrons	fr_Al_OH	fr_ketone
EState_VSA11	PEOE_VSA1	fr_Al_OH_noTert	fr_ketone_Topliss
EState_VSA2	PEOE_VSA10	fr_ArN	fr_lactam
EState_VSA3	PEOE_VSA11	fr_Ar_COO	fr_lactone
EState_VSA4	PEOE_VSA12	fr_Ar_N	fr_methoxy
EState_VSA5	PEOE_VSA13	fr_Ar_NH	fr_morpholine
EState_VSA6	PEOE_VSA14	fr_Ar_OH	fr_nitrile
EState_VSA7	PEOE_VSA2	fr_COO	fr_nitro
EState_VSA8	PEOE_VSA3	fr_COO2	fr_nitro_arom
EState_VSA9	PEOE_VSA4	fr_C_O	fr_nitro_arom_nonortho
ExactMolWt	PEOE_VSA5	fr_C_O_noCOO	fr_nitroso
FpDensityMorgan1	PEOE_VSA6	fr_C_S	fr_oxazole
FpDensityMorgan2	PEOE_VSA7	fr_HOCCN	fr_oxime
FpDensityMorgan3	PEOE_VSA8	fr_Imine	fr_para_hydroxylation
FractionCSP3	PEOE_VSA9	fr_NH0	fr_phenol
HallKierAlpha	RingCount	fr_NH1	fr_phenol_noOrthoHbond
HeavyAtomCount	SMR_VSA1	fr_NH2	fr_phos_acid
HeavyAtomMolWt	SMR_VSA10	fr_N_O	fr_phos_ester
Ipc	SMR_VSA2	fr_Ndealkylation1	fr_piperdine
Kappa1	SMR_VSA3	fr_Ndealkylation2	fr_piperzine
Kappa2	SMR_VSA4	fr_Nhpyrrole	fr_priamide
Kappa3	SMR_VSA5	fr_SH	fr_prisulfonamd
LabuteASA	SMR_VSA6	fr_aldehyde	fr_pyridine
MaxAbsEStateIndex	SMR_VSA7	fr_alkyl_carbamate	fr_quatN
MaxAbsPartialCharge	SMR_VSA8	fr_alkyl_halide	fr_sulfide
MaxEStateIndex	SMR_VSA9	fr_allylic_oxid	fr_sulfonamd
MaxPartialCharge	SlogP_VSA1	fr_amide	fr_sulfone
MinAbsEStateIndex	SlogP_VSA10	fr_amidine	fr_term_acetylene
MinAbsPartialCharge	SlogP_VSA11	fr_aniline	fr_tetrazole
MinEStateIndex	SlogP_VSA12	fr_aryl_methyl	fr_thiazole
MinPartialCharge	SlogP_VSA2	fr_azide	fr_thiocyan
MolLogP	SlogP_VSA3	fr_azo	fr_thiophene
MolMR	SlogP_VSA4	fr_barbitur	fr_unbrch_alkane
MolWt	SlogP_VSA5	fr_benzene	fr_urea
NHOHCount	SlogP_VSA6	fr_benzodiazepine	qed

Additional file 2: Scaffold folds excluded from the robustness analysis

Every scaffold fold left out of the AUC_{norm} analysis, by dataset, model, representation and noise condition, with its clean R^2. Every exclusion is a clean R^2 below 0.3; no QM9 replicate fell below it.

Table S.4: Scaffold folds excluded from the robustness analysis. One row is one model on one representation under one noise condition, in one scaffold fold, that has no AUC_{norm}. Every exclusion is a clean R^2 below 0.3, the gate below which a ratio to the clean baseline is unstable; its clean R^2 is given. All excluded folds are on the three assay datasets: no QM9 replicate fell below the gate.

Dataset	Model	Representation	Condition	Fold	Reason	Clean R^2
LogD	BNN-α (var. head)	MHG-GNN	Gaussian	1	clean R2 below the gate	0.077
LogD	BNN-α (var. head)	MHG-GNN	Grouped, wider	1	clean R2 below the gate	0.289
Caco-2	RF	ChemBERTa	Gaussian	1	clean R2 below the gate	0.292
Caco-2	RF	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.292
Caco-2	RF	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.292
Caco-2	RF	ChemBERTa	Student-t (ν=5)	1	clean R2 below the gate	0.292
Caco-2	RF	ChemBERTa	Outlier (10%)	1	clean R2 below the gate	0.292
Caco-2	RF	ChemBERTa	Laplace	1	clean R2 below the gate	0.292
Caco-2	QRF	ChemBERTa	Gaussian	1	clean R2 below the gate	0.282
Caco-2	QRF	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.282
Caco-2	QRF	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.282
Caco-2	XGBoost	ChemBERTa	Gaussian	1	clean R2 below the gate	0.276
Caco-2	XGBoost	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.276
Caco-2	XGBoost	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.276
Caco-2	NGBoost	Avalon	Gaussian	1	clean R2 below the gate	0.249
Caco-2	NGBoost	Avalon	Gaussian	3	clean R2 below the gate	0.281
Caco-2	NGBoost	Avalon	Gaussian	4	clean R2 below the gate	0.287
Caco-2	NGBoost	Avalon	Grouped, wider	1	clean R2 below the gate	0.249
Caco-2	NGBoost	Avalon	Grouped, wider	3	clean R2 below the gate	0.281
Caco-2	NGBoost	Avalon	Grouped, wider	4	clean R2 below the gate	0.287
Caco-2	NGBoost	Avalon	Grouped, shifted	1	clean R2 below the gate	0.249
Caco-2	NGBoost	Avalon	Grouped, shifted	3	clean R2 below the gate	0.281
Caco-2	NGBoost	Avalon	Grouped, shifted	4	clean R2 below the gate	0.287
Caco-2	NGBoost	ChemBERTa	Gaussian	1	clean R2 below the gate	0.202
Caco-2	NGBoost	ChemBERTa	Gaussian	3	clean R2 below the gate	0.256
Caco-2	NGBoost	ChemBERTa	Gaussian	4	clean R2 below the gate	0.243
Caco-2	NGBoost	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.202
Caco-2	NGBoost	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.256
Caco-2	NGBoost	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.243
Caco-2	NGBoost	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.202
Caco-2	NGBoost	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.256
Caco-2	NGBoost	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.243
Caco-2	NGBoost	ChemBERTa	Student-t (ν=5)	1	clean R2 below the gate	0.202
Caco-2	NGBoost	ChemBERTa	Student-t (ν=5)	3	clean R2 below the gate	0.256
Caco-2	NGBoost	ChemBERTa	Student-t (ν=5)	4	clean R2 below the gate	0.243
Caco-2	NGBoost	ChemBERTa	Outlier (10%)	1	clean R2 below the gate	0.202
Caco-2	NGBoost	ChemBERTa	Outlier (10%)	3	clean R2 below the gate	0.256
Caco-2	NGBoost	ChemBERTa	Outlier (10%)	4	clean R2 below the gate	0.243
Caco-2	NGBoost	ChemBERTa	Laplace	1	clean R2 below the gate	0.202
Caco-2	NGBoost	ChemBERTa	Laplace	3	clean R2 below the gate	0.256
Caco-2	NGBoost	ChemBERTa	Laplace	4	clean R2 below the gate	0.243
Caco-2	NGBoost	ECFP4	Gaussian	1	clean R2 below the gate	0.267
Caco-2	NGBoost	ECFP4	Gaussian	4	clean R2 below the gate	0.270
Caco-2	NGBoost	ECFP4	Grouped, wider	1	clean R2 below the gate	0.267
Caco-2	NGBoost	ECFP4	Grouped, wider	4	clean R2 below the gate	0.270
Caco-2	NGBoost	ECFP4	Grouped, shifted	1	clean R2 below the gate	0.267
Caco-2	NGBoost	ECFP4	Grouped, shifted	4	clean R2 below the gate	0.270
Caco-2	NGBoost	ECFP4	Student-t (ν=5)	1	clean R2 below the gate	0.267
Caco-2	NGBoost	ECFP4	Student-t (ν=5)	4	clean R2 below the gate	0.270
Caco-2	NGBoost	ECFP4	Outlier (10%)	1	clean R2 below the gate	0.267
Caco-2	NGBoost	ECFP4	Outlier (10%)	4	clean R2 below the gate	0.270
Caco-2	NGBoost	ECFP4	Laplace	1	clean R2 below the gate	0.267
Caco-2	NGBoost	ECFP4	Laplace	4	clean R2 below the gate	0.270
Caco-2	NGBoost	MHG-GNN	Gaussian	1	clean R2 below the gate	0.294
Caco-2	NGBoost	MHG-GNN	Grouped, wider	1	clean R2 below the gate	0.294
Caco-2	NGBoost	MHG-GNN	Grouped, shifted	1	clean R2 below the gate	0.294
Caco-2	NGBoost	Sort & Slice	Gaussian	4	clean R2 below the gate	0.296
Caco-2	NGBoost	Sort & Slice	Grouped, wider	4	clean R2 below the gate	0.296

Dataset	Model	Representation	Condition	Fold	Reason	Clean R^2
Caco-2	NGBoost	Sort & Slice	Grouped, shifted	4	clean R2 below the gate	0.296
Caco-2	BNN-α	Avalon	Gaussian	1	clean R2 below the gate	0.274
Caco-2	BNN-α	Avalon	Gaussian	3	clean R2 below the gate	0.214
Caco-2	BNN-α	Avalon	Grouped, wider	3	clean R2 below the gate	0.251
Caco-2	BNN-α	Avalon	Grouped, wider	4	clean R2 below the gate	0.247
Caco-2	BNN-α	Avalon	Grouped, shifted	4	clean R2 below the gate	0.248
Caco-2	BNN-α	ChemBERTa	Gaussian	0	clean R2 below the gate	0.297
Caco-2	BNN-α	ChemBERTa	Gaussian	1	clean R2 below the gate	0.175
Caco-2	BNN-α	ChemBERTa	Gaussian	2	clean R2 below the gate	0.202
Caco-2	BNN-α	ChemBERTa	Gaussian	3	clean R2 below the gate	0.205
Caco-2	BNN-α	ChemBERTa	Gaussian	4	clean R2 below the gate	0.144
Caco-2	BNN-α	ChemBERTa	Grouped, wider	0	clean R2 below the gate	0.230
Caco-2	BNN-α	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.207
Caco-2	BNN-α	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.181
Caco-2	BNN-α	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.144
Caco-2	BNN-α	ChemBERTa	Grouped, shifted	0	clean R2 below the gate	0.241
Caco-2	BNN-α	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.250
Caco-2	BNN-α	ChemBERTa	Grouped, shifted	2	clean R2 below the gate	0.245
Caco-2	BNN-α	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.233
Caco-2	BNN-α	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.143
Caco-2	BNN-α	MHG-GNN	Gaussian	0	clean R2 below the gate	0.130
Caco-2	BNN-α	MHG-GNN	Gaussian	1	clean R2 below the gate	0.226
Caco-2	BNN-α	MHG-GNN	Gaussian	2	clean R2 below the gate	0.273
Caco-2	BNN-α	MHG-GNN	Gaussian	3	clean R2 below the gate	0.128
Caco-2	BNN-α	MHG-GNN	Grouped, wider	1	clean R2 below the gate	0.228
Caco-2	BNN-α	MHG-GNN	Grouped, wider	2	clean R2 below the gate	0.191
Caco-2	BNN-α	MHG-GNN	Grouped, wider	3	clean R2 below the gate	0.188
Caco-2	BNN-α	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.232
Caco-2	BNN-α	MHG-GNN	Grouped, shifted	0	clean R2 below the gate	0.240
Caco-2	BNN-α	MHG-GNN	Grouped, shifted	1	clean R2 below the gate	0.091
Caco-2	BNN-α	MHG-GNN	Grouped, shifted	2	clean R2 below the gate	0.266
Caco-2	BNN-α	MHG-GNN	Grouped, shifted	3	clean R2 below the gate	0.223
Caco-2	BNN-α	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	0.132
Caco-2	BNN-α	PDV	Gaussian	3	clean R2 below the gate	0.282
Caco-2	BNN-α	PDV	Gaussian	4	clean R2 below the gate	0.290
Caco-2	BNN-α	Sort & Slice	Grouped, wider	1	clean R2 below the gate	0.292
Caco-2	BNN-α (var. head)	Avalon	Gaussian	0	clean R2 below the gate	0.296
Caco-2	BNN-α (var. head)	Avalon	Gaussian	1	clean R2 below the gate	0.258
Caco-2	BNN-α (var. head)	Avalon	Gaussian	3	clean R2 below the gate	0.258
Caco-2	BNN-α (var. head)	Avalon	Gaussian	4	clean R2 below the gate	0.132
Caco-2	BNN-α (var. head)	Avalon	Grouped, wider	1	clean R2 below the gate	0.289
Caco-2	BNN-α (var. head)	Avalon	Grouped, shifted	3	clean R2 below the gate	0.170
Caco-2	BNN-α (var. head)	ChemBERTa	Gaussian	0	clean R2 below the gate	0.229
Caco-2	BNN-α (var. head)	ChemBERTa	Gaussian	1	clean R2 below the gate	0.103
Caco-2	BNN-α (var. head)	ChemBERTa	Gaussian	3	clean R2 below the gate	0.112
Caco-2	BNN-α (var. head)	ChemBERTa	Gaussian	4	clean R2 below the gate	0.085
Caco-2	BNN-α (var. head)	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.144
Caco-2	BNN-α (var. head)	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.143
Caco-2	BNN-α (var. head)	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.293
Caco-2	BNN-α (var. head)	ChemBERTa	Grouped, shifted	0	clean R2 below the gate	0.203
Caco-2	BNN-α (var. head)	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.225
Caco-2	BNN-α (var. head)	ChemBERTa	Grouped, shifted	2	clean R2 below the gate	0.128
Caco-2	BNN-α (var. head)	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.114
Caco-2	BNN-α (var. head)	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.106
Caco-2	BNN-α (var. head)	ChemBERTa	Student-t (ν=5)	1	clean R2 below the gate	0.246
Caco-2	BNN-α (var. head)	ChemBERTa	Student-t (ν=5)	2	clean R2 below the gate	0.177
Caco-2	BNN-α (var. head)	ChemBERTa	Student-t (ν=5)	3	clean R2 below the gate	0.115
Caco-2	BNN-α (var. head)	ChemBERTa	Student-t (ν=5)	4	clean R2 below the gate	0.240
Caco-2	BNN-α (var. head)	ChemBERTa	Outlier (10%)	1	clean R2 below the gate	0.154
Caco-2	BNN-α (var. head)	ChemBERTa	Outlier (10%)	2	clean R2 below the gate	0.212
Caco-2	BNN-α (var. head)	ChemBERTa	Outlier (10%)	3	clean R2 below the gate	0.262
Caco-2	BNN-α (var. head)	ChemBERTa	Outlier (10%)	4	clean R2 below the gate	0.113
Caco-2	BNN-α (var. head)	ChemBERTa	Laplace	0	clean R2 below the gate	0.204
Caco-2	BNN-α (var. head)	ChemBERTa	Laplace	1	clean R2 below the gate	0.224
Caco-2	BNN-α (var. head)	ChemBERTa	Laplace	2	clean R2 below the gate	0.157
Caco-2	BNN-α (var. head)	ChemBERTa	Laplace	3	clean R2 below the gate	0.074
Caco-2	BNN-α (var. head)	ChemBERTa	Laplace	4	clean R2 below the gate	0.093
Caco-2	BNN-α (var. head)	ECFP4	Laplace	3	clean R2 below the gate	0.256
Caco-2	BNN-α (var. head)	MHG-GNN	Gaussian	0	clean R2 below the gate	-0.065
Caco-2	BNN-α (var. head)	MHG-GNN	Gaussian	1	clean R2 below the gate	0.057
Caco-2	BNN-α (var. head)	MHG-GNN	Gaussian	2	clean R2 below the gate	0.126
Caco-2	BNN-α (var. head)	MHG-GNN	Gaussian	3	clean R2 below the gate	0.088

Dataset	Model	Representation	Condition	Fold	Reason	Clean R^2
Caco-2	BNN-α (var. head)	MHG-GNN	Gaussian	4	clean R2 below the gate	0.103
Caco-2	BNN-α (var. head)	MHG-GNN	Grouped, wider	1	clean R2 below the gate	0.060
Caco-2	BNN-α (var. head)	MHG-GNN	Grouped, wider	2	clean R2 below the gate	0.092
Caco-2	BNN-α (var. head)	MHG-GNN	Grouped, wider	3	clean R2 below the gate	0.249
Caco-2	BNN-α (var. head)	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.043
Caco-2	BNN-α (var. head)	MHG-GNN	Grouped, shifted	0	clean R2 below the gate	0.120
Caco-2	BNN-α (var. head)	MHG-GNN	Grouped, shifted	1	clean R2 below the gate	0.019
Caco-2	BNN-α (var. head)	MHG-GNN	Grouped, shifted	3	clean R2 below the gate	0.069
Caco-2	BNN-α (var. head)	PDV	Gaussian	4	clean R2 below the gate	0.289
Caco-2	BNN-α (var. head)	PDV	Grouped, wider	4	clean R2 below the gate	0.226
Caco-2	BNN-α (var. head)	PDV	Grouped, shifted	4	clean R2 below the gate	0.210
Caco-2	BNN-α (var. head)	PDV	Student-t (ν=5)	4	clean R2 below the gate	0.275
Caco-2	BNN-α (var. head)	PDV	Outlier (10%)	1	clean R2 below the gate	0.251
Caco-2	BNN-α (var. head)	PDV	Outlier (10%)	3	clean R2 below the gate	0.297
Caco-2	BNN-α (var. head)	PDV	Laplace	0	clean R2 below the gate	0.273
Caco-2	VBLL-α	Avalon	Gaussian	0	clean R2 below the gate	0.292
Caco-2	VBLL-α	Avalon	Gaussian	1	clean R2 below the gate	0.212
Caco-2	VBLL-α	Avalon	Gaussian	3	clean R2 below the gate	0.191
Caco-2	VBLL-α	Avalon	Gaussian	4	clean R2 below the gate	0.212
Caco-2	VBLL-α	Avalon	Grouped, wider	0	clean R2 below the gate	0.268
Caco-2	VBLL-α	Avalon	Grouped, wider	1	clean R2 below the gate	0.177
Caco-2	VBLL-α	Avalon	Grouped, wider	3	clean R2 below the gate	0.194
Caco-2	VBLL-α	Avalon	Grouped, wider	4	clean R2 below the gate	0.201
Caco-2	VBLL-α	Avalon	Grouped, shifted	0	clean R2 below the gate	0.231
Caco-2	VBLL-α	Avalon	Grouped, shifted	1	clean R2 below the gate	0.239
Caco-2	VBLL-α	Avalon	Grouped, shifted	3	clean R2 below the gate	0.146
Caco-2	VBLL-α	Avalon	Grouped, shifted	4	clean R2 below the gate	0.182
Caco-2	VBLL-α	ChemBERTa	Gaussian	0	clean R2 below the gate	0.193
Caco-2	VBLL-α	ChemBERTa	Gaussian	1	clean R2 below the gate	0.249
Caco-2	VBLL-α	ChemBERTa	Gaussian	2	clean R2 below the gate	0.278
Caco-2	VBLL-α	ChemBERTa	Gaussian	3	clean R2 below the gate	0.214
Caco-2	VBLL-α	ChemBERTa	Gaussian	4	clean R2 below the gate	0.238
Caco-2	VBLL-α	ChemBERTa	Grouped, wider	0	clean R2 below the gate	0.247
Caco-2	VBLL-α	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.237
Caco-2	VBLL-α	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.214
Caco-2	VBLL-α	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.208
Caco-2	VBLL-α	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.258
Caco-2	VBLL-α	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.210
Caco-2	VBLL-α	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.213
Caco-2	VBLL-α	ECFP4	Gaussian	3	clean R2 below the gate	0.267
Caco-2	VBLL-α	ECFP4	Gaussian	4	clean R2 below the gate	0.299
Caco-2	VBLL-α	ECFP4	Grouped, wider	1	clean R2 below the gate	0.254
Caco-2	VBLL-α	ECFP4	Grouped, wider	3	clean R2 below the gate	0.290
Caco-2	VBLL-α	ECFP4	Grouped, shifted	3	clean R2 below the gate	0.260
Caco-2	VBLL-α	ECFP4	Grouped, shifted	4	clean R2 below the gate	0.272
Caco-2	VBLL-α	MHG-GNN	Gaussian	0	clean R2 below the gate	0.242
Caco-2	VBLL-α	MHG-GNN	Gaussian	1	clean R2 below the gate	0.211
Caco-2	VBLL-α	MHG-GNN	Gaussian	2	clean R2 below the gate	0.285
Caco-2	VBLL-α	MHG-GNN	Gaussian	3	clean R2 below the gate	0.219
Caco-2	VBLL-α	MHG-GNN	Gaussian	4	clean R2 below the gate	0.159
Caco-2	VBLL-α	MHG-GNN	Grouped, wider	0	clean R2 below the gate	0.267
Caco-2	VBLL-α	MHG-GNN	Grouped, wider	1	clean R2 below the gate	0.268
Caco-2	VBLL-α	MHG-GNN	Grouped, wider	3	clean R2 below the gate	0.189
Caco-2	VBLL-α	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.186
Caco-2	VBLL-α	MHG-GNN	Grouped, shifted	1	clean R2 below the gate	0.242
Caco-2	VBLL-α	MHG-GNN	Grouped, shifted	3	clean R2 below the gate	0.203
Caco-2	VBLL-α	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	0.227
Caco-2	VBLL-α	PDV	Gaussian	4	clean R2 below the gate	0.299
Caco-2	VBLL-α	PDV	Grouped, wider	4	clean R2 below the gate	0.269
Caco-2	VBLL-α	PDV	Grouped, shifted	3	clean R2 below the gate	0.295
Caco-2	VBLL-α	PDV	Grouped, shifted	4	clean R2 below the gate	0.292
Caco-2	VBLL-α	Sort & Slice	Gaussian	3	clean R2 below the gate	0.297
Caco-2	VBLL-α	Sort & Slice	Grouped, wider	3	clean R2 below the gate	0.237
Caco-2	VBLL-α	Sort & Slice	Grouped, shifted	3	clean R2 below the gate	0.277
Caco-2	VBLL-α	Sort & Slice	Grouped, shifted	4	clean R2 below the gate	0.297
Caco-2	VBLL-α (het.)	Avalon	Gaussian	1	clean R2 below the gate	0.246
Caco-2	VBLL-α (het.)	Avalon	Gaussian	3	clean R2 below the gate	0.214
Caco-2	VBLL-α (het.)	Avalon	Gaussian	4	clean R2 below the gate	0.210
Caco-2	VBLL-α (het.)	Avalon	Grouped, wider	1	clean R2 below the gate	0.259
Caco-2	VBLL-α (het.)	Avalon	Grouped, wider	3	clean R2 below the gate	0.166
Caco-2	VBLL-α (het.)	Avalon	Grouped, wider	4	clean R2 below the gate	0.231
Caco-2	VBLL-α (het.)	Avalon	Grouped, shifted	1	clean R2 below the gate	0.259

Dataset	Model	Representation	Condition	Fold	Reason	Clean R^2
Caco-2	VBLL-α (het.)	Avalon	Grouped, shifted	3	clean R2 below the gate	0.235
Caco-2	VBLL-α (het.)	Avalon	Grouped, shifted	4	clean R2 below the gate	0.203
Caco-2	VBLL-α (het.)	ChemBERTa	Gaussian	1	clean R2 below the gate	0.249
Caco-2	VBLL-α (het.)	ChemBERTa	Gaussian	3	clean R2 below the gate	0.243
Caco-2	VBLL-α (het.)	ChemBERTa	Gaussian	4	clean R2 below the gate	0.253
Caco-2	VBLL-α (het.)	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.281
Caco-2	VBLL-α (het.)	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.192
Caco-2	VBLL-α (het.)	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.282
Caco-2	VBLL-α (het.)	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.250
Caco-2	VBLL-α (het.)	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.265
Caco-2	VBLL-α (het.)	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.272
Caco-2	VBLL-α (het.)	ECFP4	Gaussian	3	clean R2 below the gate	0.242
Caco-2	VBLL-α (het.)	ECFP4	Grouped, wider	3	clean R2 below the gate	0.282
Caco-2	VBLL-α (het.)	ECFP4	Grouped, shifted	3	clean R2 below the gate	0.294
Caco-2	VBLL-α (het.)	ECFP4	Grouped, shifted	4	clean R2 below the gate	0.293
Caco-2	VBLL-α (het.)	MHG-GNN	Gaussian	3	clean R2 below the gate	0.236
Caco-2	VBLL-α (het.)	MHG-GNN	Gaussian	4	clean R2 below the gate	0.261
Caco-2	VBLL-α (het.)	MHG-GNN	Grouped, wider	3	clean R2 below the gate	0.221
Caco-2	VBLL-α (het.)	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.280
Caco-2	VBLL-α (het.)	MHG-GNN	Grouped, shifted	1	clean R2 below the gate	0.294
Caco-2	VBLL-α (het.)	MHG-GNN	Grouped, shifted	3	clean R2 below the gate	0.220
Caco-2	VBLL-α (het.)	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	0.252
Caco-2	VBLL-α (het.)	PDV	Grouped, shifted	4	clean R2 below the gate	0.282
Caco-2	VBLL-α (het.)	Sort & Slice	Gaussian	3	clean R2 below the gate	0.293
Caco-2	BNN-β	Avalon	Gaussian	3	clean R2 below the gate	0.121
Caco-2	BNN-β	Avalon	Gaussian	4	clean R2 below the gate	0.298
Caco-2	BNN-β	Avalon	Grouped, wider	3	clean R2 below the gate	0.271
Caco-2	BNN-β	Avalon	Grouped, wider	4	clean R2 below the gate	0.276
Caco-2	BNN-β	Avalon	Grouped, shifted	0	clean R2 below the gate	0.256
Caco-2	BNN-β	Avalon	Grouped, shifted	3	clean R2 below the gate	0.186
Caco-2	BNN-β	ChemBERTa	Gaussian	1	clean R2 below the gate	0.187
Caco-2	BNN-β	ChemBERTa	Gaussian	3	clean R2 below the gate	0.126
Caco-2	BNN-β	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.190
Caco-2	BNN-β	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.209
Caco-2	BNN-β	ChemBERTa	Grouped, shifted	0	clean R2 below the gate	0.285
Caco-2	BNN-β	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.238
Caco-2	BNN-β	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.192
Caco-2	BNN-β	MHG-GNN	Gaussian	1	clean R2 below the gate	0.137
Caco-2	BNN-β	MHG-GNN	Gaussian	3	clean R2 below the gate	0.256
Caco-2	BNN-β	MHG-GNN	Grouped, wider	3	clean R2 below the gate	0.157
Caco-2	BNN-β	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.240
Caco-2	BNN-β	MHG-GNN	Grouped, shifted	1	clean R2 below the gate	0.152
Caco-2	BNN-β	MHG-GNN	Grouped, shifted	2	clean R2 below the gate	0.274
Caco-2	BNN-β	MHG-GNN	Grouped, shifted	3	clean R2 below the gate	0.239
Caco-2	BNN-β	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	0.211
Caco-2	BNN-β (var. head)	Avalon	Gaussian	2	clean R2 below the gate	0.265
Caco-2	BNN-β (var. head)	Avalon	Gaussian	3	clean R2 below the gate	0.239
Caco-2	BNN-β (var. head)	Avalon	Gaussian	4	clean R2 below the gate	0.298
Caco-2	BNN-β (var. head)	Avalon	Grouped, wider	3	clean R2 below the gate	0.271
Caco-2	BNN-β (var. head)	Avalon	Grouped, wider	4	clean R2 below the gate	0.291
Caco-2	BNN-β (var. head)	Avalon	Grouped, shifted	1	clean R2 below the gate	0.244
Caco-2	BNN-β (var. head)	Avalon	Grouped, shifted	3	clean R2 below the gate	0.220
Caco-2	BNN-β (var. head)	Avalon	Grouped, shifted	4	clean R2 below the gate	0.249
Caco-2	BNN-β (var. head)	ChemBERTa	Gaussian	0	clean R2 below the gate	0.209
Caco-2	BNN-β (var. head)	ChemBERTa	Gaussian	1	clean R2 below the gate	0.173
Caco-2	BNN-β (var. head)	ChemBERTa	Gaussian	3	clean R2 below the gate	0.279
Caco-2	BNN-β (var. head)	ChemBERTa	Gaussian	4	clean R2 below the gate	0.124
Caco-2	BNN-β (var. head)	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.255
Caco-2	BNN-β (var. head)	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.129
Caco-2	BNN-β (var. head)	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.121
Caco-2	BNN-β (var. head)	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.088
Caco-2	BNN-β (var. head)	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.229
Caco-2	BNN-β (var. head)	MHG-GNN	Gaussian	0	clean R2 below the gate	-0.260
Caco-2	BNN-β (var. head)	MHG-GNN	Gaussian	1	clean R2 below the gate	0.145
Caco-2	BNN-β (var. head)	MHG-GNN	Gaussian	2	clean R2 below the gate	0.195
Caco-2	BNN-β (var. head)	MHG-GNN	Gaussian	3	clean R2 below the gate	0.132
Caco-2	BNN-β (var. head)	MHG-GNN	Grouped, wider	0	clean R2 below the gate	-0.280
Caco-2	BNN-β (var. head)	MHG-GNN	Grouped, wider	2	clean R2 below the gate	0.056
Caco-2	BNN-β (var. head)	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.095
Caco-2	BNN-β (var. head)	MHG-GNN	Grouped, shifted	0	clean R2 below the gate	-0.057
Caco-2	BNN-β (var. head)	MHG-GNN	Grouped, shifted	1	clean R2 below the gate	0.194
Caco-2	BNN-β (var. head)	MHG-GNN	Grouped, shifted	2	clean R2 below the gate	0.201

Dataset	Model	Representation	Condition	Fold	Reason	Clean R^2
Caco-2	BNN-β (var. head)	MHG-GNN	Grouped, shifted	3	clean R2 below the gate	0.083
Caco-2	BNN-β (var. head)	PDV	Grouped, wider	4	clean R2 below the gate	0.290
Caco-2	BNN-β (var. head)	PDV	Grouped, shifted	3	clean R2 below the gate	0.270
Caco-2	VBLL-β	Avalon	Gaussian	3	clean R2 below the gate	0.259
Caco-2	VBLL-β	Avalon	Gaussian	4	clean R2 below the gate	0.299
Caco-2	VBLL-β	Avalon	Grouped, wider	3	clean R2 below the gate	0.270
Caco-2	VBLL-β	Avalon	Grouped, wider	4	clean R2 below the gate	0.284
Caco-2	VBLL-β	Avalon	Grouped, shifted	3	clean R2 below the gate	0.159
Caco-2	VBLL-β	Avalon	Grouped, shifted	4	clean R2 below the gate	0.296
Caco-2	VBLL-β	ChemBERTa	Gaussian	3	clean R2 below the gate	0.272
Caco-2	VBLL-β	ChemBERTa	Gaussian	4	clean R2 below the gate	0.272
Caco-2	VBLL-β	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.291
Caco-2	VBLL-β	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.291
Caco-2	VBLL-β	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.283
Caco-2	VBLL-β	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.299
Caco-2	VBLL-β	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.265
Caco-2	VBLL-β	MHG-GNN	Gaussian	1	clean R2 below the gate	0.260
Caco-2	VBLL-β	MHG-GNN	Gaussian	3	clean R2 below the gate	0.272
Caco-2	VBLL-β	MHG-GNN	Gaussian	4	clean R2 below the gate	0.253
Caco-2	VBLL-β	MHG-GNN	Grouped, wider	1	clean R2 below the gate	0.256
Caco-2	VBLL-β	MHG-GNN	Grouped, wider	3	clean R2 below the gate	0.262
Caco-2	VBLL-β	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.280
Caco-2	VBLL-β	MHG-GNN	Grouped, shifted	1	clean R2 below the gate	0.294
Caco-2	VBLL-β	MHG-GNN	Grouped, shifted	3	clean R2 below the gate	0.239
Caco-2	VBLL-β	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	0.253
Caco-2	VBLL-β	PDV	Gaussian	4	clean R2 below the gate	0.272
Caco-2	VBLL-β (het.)	Avalon	Gaussian	3	clean R2 below the gate	0.259
Caco-2	VBLL-β (het.)	Avalon	Gaussian	4	clean R2 below the gate	0.257
Caco-2	VBLL-β (het.)	Avalon	Grouped, wider	0	clean R2 below the gate	0.268
Caco-2	VBLL-β (het.)	Avalon	Grouped, wider	1	clean R2 below the gate	0.281
Caco-2	VBLL-β (het.)	Avalon	Grouped, wider	3	clean R2 below the gate	0.265
Caco-2	VBLL-β (het.)	Avalon	Grouped, wider	4	clean R2 below the gate	0.281
Caco-2	VBLL-β (het.)	Avalon	Grouped, shifted	1	clean R2 below the gate	0.294
Caco-2	VBLL-β (het.)	Avalon	Grouped, shifted	3	clean R2 below the gate	0.262
Caco-2	VBLL-β (het.)	Avalon	Grouped, shifted	4	clean R2 below the gate	0.228
Caco-2	VBLL-β (het.)	ChemBERTa	Gaussian	1	clean R2 below the gate	0.253
Caco-2	VBLL-β (het.)	ChemBERTa	Gaussian	3	clean R2 below the gate	0.262
Caco-2	VBLL-β (het.)	ChemBERTa	Gaussian	4	clean R2 below the gate	0.227
Caco-2	VBLL-β (het.)	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.260
Caco-2	VBLL-β (het.)	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.294
Caco-2	VBLL-β (het.)	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.261
Caco-2	VBLL-β (het.)	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.299
Caco-2	VBLL-β (het.)	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.287
Caco-2	VBLL-β (het.)	ChemBERTa	Student-t (ν=5)	3	clean R2 below the gate	0.247
Caco-2	VBLL-β (het.)	ChemBERTa	Student-t (ν=5)	4	clean R2 below the gate	0.268
Caco-2	VBLL-β (het.)	ChemBERTa	Outlier (10%)	1	clean R2 below the gate	0.266
Caco-2	VBLL-β (het.)	ChemBERTa	Outlier (10%)	3	clean R2 below the gate	0.269
Caco-2	VBLL-β (het.)	ChemBERTa	Outlier (10%)	4	clean R2 below the gate	0.296
Caco-2	VBLL-β (het.)	ChemBERTa	Laplace	3	clean R2 below the gate	0.271
Caco-2	VBLL-β (het.)	ChemBERTa	Laplace	4	clean R2 below the gate	0.296
Caco-2	VBLL-β (het.)	ECFP4	Gaussian	3	clean R2 below the gate	0.297
Caco-2	VBLL-β (het.)	ECFP4	Outlier (10%)	3	clean R2 below the gate	0.292
Caco-2	VBLL-β (het.)	MHG-GNN	Gaussian	3	clean R2 below the gate	0.270
Caco-2	VBLL-β (het.)	MHG-GNN	Gaussian	4	clean R2 below the gate	0.282
Caco-2	VBLL-β (het.)	MHG-GNN	Grouped, wider	3	clean R2 below the gate	0.202
Caco-2	VBLL-β (het.)	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.273
Caco-2	VBLL-β (het.)	MHG-GNN	Grouped, shifted	1	clean R2 below the gate	0.149
Caco-2	VBLL-β (het.)	MHG-GNN	Grouped, shifted	3	clean R2 below the gate	0.265
hERG K_i	NGBoost	Avalon	Gaussian	4	clean R2 below the gate	0.283
hERG K_i	NGBoost	Avalon	Grouped, wider	4	clean R2 below the gate	0.283
hERG K_i	NGBoost	Avalon	Grouped, shifted	4	clean R2 below the gate	0.283
hERG K_i	NGBoost	ChemBERTa	Gaussian	4	clean R2 below the gate	0.247
hERG K_i	NGBoost	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.247
hERG K_i	NGBoost	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.247
hERG K_i	NGBoost	ChemBERTa	Student-t (ν=5)	4	clean R2 below the gate	0.247
hERG K_i	NGBoost	ChemBERTa	Outlier (10%)	4	clean R2 below the gate	0.247
hERG K_i	NGBoost	ChemBERTa	Laplace	4	clean R2 below the gate	0.247
hERG K_i	NGBoost	PDV	Gaussian	4	clean R2 below the gate	0.297
hERG K_i	NGBoost	PDV	Grouped, wider	4	clean R2 below the gate	0.297
hERG K_i	NGBoost	PDV	Grouped, shifted	4	clean R2 below the gate	0.297
hERG K_i	NGBoost	PDV	Censoring	4	clean R2 below the gate	0.297
hERG K_i	NGBoost	PDV	Student-t (ν=5)	4	clean R2 below the gate	0.297

Dataset	Model	Representation	Condition	Fold	Reason	Clean R^2
hERG K_i	NGBoost	PDV	Outlier (10%)	4	clean R2 below the gate	0.297
hERG K_i	NGBoost	PDV	Laplace	4	clean R2 below the gate	0.297
hERG K_i	NN-α	PDV	Grouped, shifted	2	clean R2 below the gate	0.233
hERG K_i	NN-α	Sort & Slice	Grouped, wider	3	clean R2 below the gate	0.265
hERG K_i	BNN-α	Avalon	Grouped, wider	3	clean R2 below the gate	0.261
hERG K_i	BNN-α	ChemBERTa	Gaussian	0	clean R2 below the gate	0.072
hERG K_i	BNN-α	ChemBERTa	Gaussian	1	clean R2 below the gate	0.231
hERG K_i	BNN-α	ChemBERTa	Gaussian	2	clean R2 below the gate	0.067
hERG K_i	BNN-α	ChemBERTa	Gaussian	3	clean R2 below the gate	0.243
hERG K_i	BNN-α	ChemBERTa	Gaussian	4	clean R2 below the gate	0.194
hERG K_i	BNN-α	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.137
hERG K_i	BNN-α	ChemBERTa	Grouped, wider	2	clean R2 below the gate	0.152
hERG K_i	BNN-α	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.257
hERG K_i	BNN-α	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.050
hERG K_i	BNN-α	ChemBERTa	Grouped, shifted	0	clean R2 below the gate	0.061
hERG K_i	BNN-α	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.112
hERG K_i	BNN-α	ChemBERTa	Grouped, shifted	2	clean R2 below the gate	0.109
hERG K_i	BNN-α	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.258
hERG K_i	BNN-α	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.218
hERG K_i	BNN-α	MHG-GNN	Gaussian	0	clean R2 below the gate	0.242
hERG K_i	BNN-α	MHG-GNN	Gaussian	2	clean R2 below the gate	0.031
hERG K_i	BNN-α	MHG-GNN	Gaussian	3	clean R2 below the gate	0.127
hERG K_i	BNN-α	MHG-GNN	Gaussian	4	clean R2 below the gate	0.032
hERG K_i	BNN-α	MHG-GNN	Grouped, wider	2	clean R2 below the gate	0.051
hERG K_i	BNN-α	MHG-GNN	Grouped, wider	3	clean R2 below the gate	0.109
hERG K_i	BNN-α	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.054
hERG K_i	BNN-α	MHG-GNN	Grouped, shifted	2	clean R2 below the gate	0.079
hERG K_i	BNN-α	MHG-GNN	Grouped, shifted	3	clean R2 below the gate	0.181
hERG K_i	BNN-α	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	0.138
hERG K_i	BNN-α	PDV	Gaussian	2	clean R2 below the gate	0.154
hERG K_i	BNN-α	PDV	Gaussian	4	clean R2 below the gate	0.173
hERG K_i	BNN-α	PDV	Grouped, wider	2	clean R2 below the gate	0.132
hERG K_i	BNN-α	PDV	Grouped, wider	4	clean R2 below the gate	0.153
hERG K_i	BNN-α	PDV	Grouped, shifted	4	clean R2 below the gate	0.097
hERG K_i	BNN-α	Sort & Slice	Gaussian	2	clean R2 below the gate	0.037
hERG K_i	BNN-α (var. head)	Avalon	Gaussian	0	clean R2 below the gate	0.141
hERG K_i	BNN-α (var. head)	Avalon	Gaussian	4	clean R2 below the gate	0.223
hERG K_i	BNN-α (var. head)	ChemBERTa	Gaussian	0	clean R2 below the gate	-0.011
hERG K_i	BNN-α (var. head)	ChemBERTa	Gaussian	1	clean R2 below the gate	0.060
hERG K_i	BNN-α (var. head)	ChemBERTa	Gaussian	2	clean R2 below the gate	0.011
hERG K_i	BNN-α (var. head)	ChemBERTa	Gaussian	3	clean R2 below the gate	0.279
hERG K_i	BNN-α (var. head)	ChemBERTa	Gaussian	4	clean R2 below the gate	0.152
hERG K_i	BNN-α (var. head)	ChemBERTa	Grouped, wider	0	clean R2 below the gate	-0.003
hERG K_i	BNN-α (var. head)	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.031
hERG K_i	BNN-α (var. head)	ChemBERTa	Grouped, wider	2	clean R2 below the gate	0.027
hERG K_i	BNN-α (var. head)	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.152
hERG K_i	BNN-α (var. head)	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.061
hERG K_i	BNN-α (var. head)	ChemBERTa	Grouped, shifted	0	clean R2 below the gate	0.055
hERG K_i	BNN-α (var. head)	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.011
hERG K_i	BNN-α (var. head)	ChemBERTa	Grouped, shifted	2	clean R2 below the gate	0.007
hERG K_i	BNN-α (var. head)	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.093
hERG K_i	BNN-α (var. head)	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.015
hERG K_i	BNN-α (var. head)	ChemBERTa	Student-t (ν=5)	0	clean R2 below the gate	-0.066
hERG K_i	BNN-α (var. head)	ChemBERTa	Student-t (ν=5)	1	clean R2 below the gate	0.119
hERG K_i	BNN-α (var. head)	ChemBERTa	Student-t (ν=5)	2	clean R2 below the gate	0.240
hERG K_i	BNN-α (var. head)	ChemBERTa	Student-t (ν=5)	3	clean R2 below the gate	0.066
hERG K_i	BNN-α (var. head)	ChemBERTa	Student-t (ν=5)	4	clean R2 below the gate	0.036
hERG K_i	BNN-α (var. head)	ChemBERTa	Outlier (10%)	0	clean R2 below the gate	0.016
hERG K_i	BNN-α (var. head)	ChemBERTa	Outlier (10%)	1	clean R2 below the gate	0.054
hERG K_i	BNN-α (var. head)	ChemBERTa	Outlier (10%)	2	clean R2 below the gate	0.032
hERG K_i	BNN-α (var. head)	ChemBERTa	Outlier (10%)	3	clean R2 below the gate	0.181
hERG K_i	BNN-α (var. head)	ChemBERTa	Outlier (10%)	4	clean R2 below the gate	0.036
hERG K_i	BNN-α (var. head)	ChemBERTa	Laplace	0	clean R2 below the gate	0.283
hERG K_i	BNN-α (var. head)	ChemBERTa	Laplace	1	clean R2 below the gate	0.053
hERG K_i	BNN-α (var. head)	ChemBERTa	Laplace	2	clean R2 below the gate	0.076
hERG K_i	BNN-α (var. head)	ChemBERTa	Laplace	3	clean R2 below the gate	0.031
hERG K_i	BNN-α (var. head)	ChemBERTa	Laplace	4	clean R2 below the gate	0.012
hERG K_i	BNN-α (var. head)	MHG-GNN	Gaussian	1	clean R2 below the gate	0.093
hERG K_i	BNN-α (var. head)	MHG-GNN	Gaussian	2	clean R2 below the gate	-0.047
hERG K_i	BNN-α (var. head)	MHG-GNN	Gaussian	4	clean R2 below the gate	0.023
hERG K_i	BNN-α (var. head)	MHG-GNN	Grouped, wider	0	clean R2 below the gate	-0.001
hERG K_i	BNN-α (var. head)	MHG-GNN	Grouped, wider	1	clean R2 below the gate	0.007

Dataset	Model	Representation	Condition	Fold	Reason	Clean R^2
hERG K_i	BNN-α (var. head)	MHG-GNN	Grouped, wider	2	clean R2 below the gate	-0.010
hERG K_i	BNN-α (var. head)	MHG-GNN	Grouped, wider	3	clean R2 below the gate	0.041
hERG K_i	BNN-α (var. head)	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.033
hERG K_i	BNN-α (var. head)	MHG-GNN	Grouped, shifted	1	clean R2 below the gate	0.083
hERG K_i	BNN-α (var. head)	MHG-GNN	Grouped, shifted	2	clean R2 below the gate	0.033
hERG K_i	BNN-α (var. head)	MHG-GNN	Grouped, shifted	3	clean R2 below the gate	-0.426
hERG K_i	BNN-α (var. head)	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	-0.035
hERG K_i	BNN-α (var. head)	PDV	Gaussian	2	clean R2 below the gate	0.075
hERG K_i	BNN-α (var. head)	PDV	Gaussian	3	clean R2 below the gate	0.169
hERG K_i	BNN-α (var. head)	PDV	Gaussian	4	clean R2 below the gate	0.259
hERG K_i	BNN-α (var. head)	PDV	Grouped, wider	1	clean R2 below the gate	0.077
hERG K_i	BNN-α (var. head)	PDV	Grouped, wider	2	clean R2 below the gate	0.210
hERG K_i	BNN-α (var. head)	PDV	Grouped, wider	3	clean R2 below the gate	0.098
hERG K_i	BNN-α (var. head)	PDV	Grouped, wider	4	clean R2 below the gate	0.217
hERG K_i	BNN-α (var. head)	PDV	Grouped, shifted	0	clean R2 below the gate	0.051
hERG K_i	BNN-α (var. head)	PDV	Grouped, shifted	2	clean R2 below the gate	0.036
hERG K_i	BNN-α (var. head)	PDV	Grouped, shifted	3	clean R2 below the gate	0.296
hERG K_i	BNN-α (var. head)	PDV	Grouped, shifted	4	clean R2 below the gate	0.153
hERG K_i	BNN-α (var. head)	PDV	Censoring	0	clean R2 below the gate	0.016
hERG K_i	BNN-α (var. head)	PDV	Censoring	1	clean R2 below the gate	0.110
hERG K_i	BNN-α (var. head)	PDV	Censoring	2	clean R2 below the gate	0.019
hERG K_i	BNN-α (var. head)	PDV	Censoring	3	clean R2 below the gate	0.219
hERG K_i	BNN-α (var. head)	PDV	Censoring	4	clean R2 below the gate	0.194
hERG K_i	BNN-α (var. head)	PDV	Student-t (ν=5)	1	clean R2 below the gate	0.074
hERG K_i	BNN-α (var. head)	PDV	Student-t (ν=5)	2	clean R2 below the gate	0.031
hERG K_i	BNN-α (var. head)	PDV	Student-t (ν=5)	3	clean R2 below the gate	0.281
hERG K_i	BNN-α (var. head)	PDV	Student-t (ν=5)	4	clean R2 below the gate	0.084
hERG K_i	BNN-α (var. head)	PDV	Outlier (10%)	0	clean R2 below the gate	0.158
hERG K_i	BNN-α (var. head)	PDV	Outlier (10%)	1	clean R2 below the gate	0.152
hERG K_i	BNN-α (var. head)	PDV	Outlier (10%)	2	clean R2 below the gate	0.213
hERG K_i	BNN-α (var. head)	PDV	Outlier (10%)	3	clean R2 below the gate	0.110
hERG K_i	BNN-α (var. head)	PDV	Outlier (10%)	4	clean R2 below the gate	0.015
hERG K_i	BNN-α (var. head)	PDV	Laplace	0	clean R2 below the gate	0.289
hERG K_i	BNN-α (var. head)	PDV	Laplace	2	clean R2 below the gate	0.113
hERG K_i	BNN-α (var. head)	PDV	Laplace	3	clean R2 below the gate	0.088
hERG K_i	BNN-α (var. head)	PDV	Laplace	4	clean R2 below the gate	0.213
hERG K_i	BNN-α (var. head)	Sort & Slice	Gaussian	1	clean R2 below the gate	0.070
hERG K_i	BNN-α (var. head)	Sort & Slice	Gaussian	2	clean R2 below the gate	0.033
hERG K_i	BNN-α (var. head)	Sort & Slice	Gaussian	3	clean R2 below the gate	0.288
hERG K_i	BNN-α (var. head)	Sort & Slice	Grouped, wider	4	clean R2 below the gate	0.201
hERG K_i	BNN-α (var. head)	Sort & Slice	Grouped, shifted	4	clean R2 below the gate	0.222
hERG K_i	VBLL-α	Avalon	Gaussian	4	clean R2 below the gate	0.275
hERG K_i	VBLL-α	Avalon	Grouped, shifted	4	clean R2 below the gate	0.299
hERG K_i	VBLL-α	ChemBERTa	Gaussian	2	clean R2 below the gate	0.293
hERG K_i	VBLL-α	ChemBERTa	Gaussian	4	clean R2 below the gate	0.221
hERG K_i	VBLL-α	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.190
hERG K_i	VBLL-α	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.287
hERG K_i	VBLL-α	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.224
hERG K_i	VBLL-α	ECFP4	Gaussian	4	clean R2 below the gate	0.288
hERG K_i	VBLL-α	ECFP4	Grouped, shifted	4	clean R2 below the gate	0.290
hERG K_i	VBLL-α	MHG-GNN	Gaussian	4	clean R2 below the gate	0.198
hERG K_i	VBLL-α	MHG-GNN	Grouped, wider	3	clean R2 below the gate	0.289
hERG K_i	VBLL-α	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.222
hERG K_i	VBLL-α	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	0.224
hERG K_i	VBLL-α	PDV	Gaussian	4	clean R2 below the gate	0.234
hERG K_i	VBLL-α	PDV	Grouped, wider	4	clean R2 below the gate	0.267
hERG K_i	VBLL-α	PDV	Grouped, shifted	4	clean R2 below the gate	0.243
hERG K_i	VBLL-α (het.)	Avalon	Grouped, wider	4	clean R2 below the gate	0.300
hERG K_i	VBLL-α (het.)	Avalon	Grouped, shifted	4	clean R2 below the gate	0.249
hERG K_i	VBLL-α (het.)	ChemBERTa	Gaussian	0	clean R2 below the gate	0.246
hERG K_i	VBLL-α (het.)	ChemBERTa	Gaussian	2	clean R2 below the gate	0.227
hERG K_i	VBLL-α (het.)	ChemBERTa	Gaussian	3	clean R2 below the gate	0.176
hERG K_i	VBLL-α (het.)	ChemBERTa	Gaussian	4	clean R2 below the gate	0.184
hERG K_i	VBLL-α (het.)	ChemBERTa	Grouped, wider	0	clean R2 below the gate	0.277
hERG K_i	VBLL-α (het.)	ChemBERTa	Grouped, wider	2	clean R2 below the gate	0.230
hERG K_i	VBLL-α (het.)	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.253
hERG K_i	VBLL-α (het.)	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.191
hERG K_i	VBLL-α (het.)	ChemBERTa	Grouped, shifted	0	clean R2 below the gate	0.241
hERG K_i	VBLL-α (het.)	ChemBERTa	Grouped, shifted	2	clean R2 below the gate	0.208
hERG K_i	VBLL-α (het.)	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.271
hERG K_i	VBLL-α (het.)	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.202
hERG K_i	VBLL-α (het.)	ECFP4	Gaussian	4	clean R2 below the gate	0.245

Dataset	Model	Representation	Condition	Fold	Reason	Clean R^2
hERG K_i	VBLL-α (het.)	ECFP4	Grouped, wider	4	clean R2 below the gate	0.298
hERG K_i	VBLL-α (het.)	ECFP4	Grouped, shifted	4	clean R2 below the gate	0.261
hERG K_i	VBLL-α (het.)	MHG-GNN	Gaussian	0	clean R2 below the gate	0.297
hERG K_i	VBLL-α (het.)	MHG-GNN	Gaussian	1	clean R2 below the gate	0.289
hERG K_i	VBLL-α (het.)	MHG-GNN	Gaussian	2	clean R2 below the gate	0.260
hERG K_i	VBLL-α (het.)	MHG-GNN	Gaussian	3	clean R2 below the gate	0.281
hERG K_i	VBLL-α (het.)	MHG-GNN	Gaussian	4	clean R2 below the gate	0.165
hERG K_i	VBLL-α (het.)	MHG-GNN	Grouped, wider	0	clean R2 below the gate	0.286
hERG K_i	VBLL-α (het.)	MHG-GNN	Grouped, wider	2	clean R2 below the gate	0.219
hERG K_i	VBLL-α (het.)	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.202
hERG K_i	VBLL-α (het.)	MHG-GNN	Grouped, shifted	2	clean R2 below the gate	0.197
hERG K_i	VBLL-α (het.)	MHG-GNN	Grouped, shifted	3	clean R2 below the gate	0.285
hERG K_i	VBLL-α (het.)	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	0.175
hERG K_i	VBLL-α (het.)	PDV	Gaussian	2	clean R2 below the gate	0.280
hERG K_i	VBLL-α (het.)	PDV	Gaussian	3	clean R2 below the gate	0.276
hERG K_i	VBLL-α (het.)	PDV	Gaussian	4	clean R2 below the gate	0.189
hERG K_i	VBLL-α (het.)	PDV	Grouped, wider	2	clean R2 below the gate	0.264
hERG K_i	VBLL-α (het.)	PDV	Grouped, wider	4	clean R2 below the gate	0.188
hERG K_i	VBLL-α (het.)	PDV	Grouped, shifted	2	clean R2 below the gate	0.235
hERG K_i	VBLL-α (het.)	PDV	Grouped, shifted	3	clean R2 below the gate	0.209
hERG K_i	VBLL-α (het.)	PDV	Grouped, shifted	4	clean R2 below the gate	0.139
hERG K_i	VBLL-α (het.)	Sort & Slice	Gaussian	4	clean R2 below the gate	0.282
hERG K_i	VBLL-α (het.)	Sort & Slice	Grouped, shifted	4	clean R2 below the gate	0.289
hERG K_i	NN-β	PDV	Grouped, wider	3	clean R2 below the gate	0.221
hERG K_i	NN-β	PDV	Grouped, shifted	3	clean R2 below the gate	-0.092
hERG K_i	BNN-β	Avalon	Gaussian	2	clean R2 below the gate	0.271
hERG K_i	BNN-β	Avalon	Gaussian	3	clean R2 below the gate	0.209
hERG K_i	BNN-β	Avalon	Grouped, shifted	4	clean R2 below the gate	0.298
hERG K_i	BNN-β	ChemBERTa	Gaussian	3	clean R2 below the gate	0.221
hERG K_i	BNN-β	ChemBERTa	Gaussian	4	clean R2 below the gate	0.210
hERG K_i	BNN-β	ChemBERTa	Grouped, wider	2	clean R2 below the gate	0.124
hERG K_i	BNN-β	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.296
hERG K_i	BNN-β	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.217
hERG K_i	BNN-β	ChemBERTa	Grouped, shifted	0	clean R2 below the gate	0.063
hERG K_i	BNN-β	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.148
hERG K_i	BNN-β	ChemBERTa	Grouped, shifted	2	clean R2 below the gate	0.272
hERG K_i	BNN-β	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.180
hERG K_i	BNN-β	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.201
hERG K_i	BNN-β	MHG-GNN	Gaussian	0	clean R2 below the gate	0.202
hERG K_i	BNN-β	MHG-GNN	Gaussian	2	clean R2 below the gate	0.084
hERG K_i	BNN-β	MHG-GNN	Gaussian	3	clean R2 below the gate	0.285
hERG K_i	BNN-β	MHG-GNN	Gaussian	4	clean R2 below the gate	0.181
hERG K_i	BNN-β	MHG-GNN	Grouped, wider	2	clean R2 below the gate	0.196
hERG K_i	BNN-β	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.074
hERG K_i	BNN-β	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	0.062
hERG K_i	BNN-β	PDV	Gaussian	2	clean R2 below the gate	0.257
hERG K_i	BNN-β	PDV	Gaussian	4	clean R2 below the gate	0.207
hERG K_i	BNN-β	PDV	Grouped, wider	2	clean R2 below the gate	0.156
hERG K_i	BNN-β	PDV	Grouped, wider	4	clean R2 below the gate	0.249
hERG K_i	BNN-β	PDV	Grouped, shifted	1	clean R2 below the gate	0.269
hERG K_i	BNN-β	PDV	Grouped, shifted	2	clean R2 below the gate	0.181
hERG K_i	BNN-β	PDV	Grouped, shifted	4	clean R2 below the gate	0.197
hERG K_i	BNN-β	Sort & Slice	Gaussian	4	clean R2 below the gate	0.063
hERG K_i	BNN-β	Sort & Slice	Grouped, shifted	1	clean R2 below the gate	0.198
hERG K_i	BNN-β	Sort & Slice	Grouped, shifted	3	clean R2 below the gate	0.074
hERG K_i	BNN-β (var. head)	Avalon	Grouped, wider	4	clean R2 below the gate	0.134
hERG K_i	BNN-β (var. head)	ChemBERTa	Gaussian	0	clean R2 below the gate	-0.015
hERG K_i	BNN-β (var. head)	ChemBERTa	Gaussian	1	clean R2 below the gate	0.191
hERG K_i	BNN-β (var. head)	ChemBERTa	Gaussian	2	clean R2 below the gate	0.248
hERG K_i	BNN-β (var. head)	ChemBERTa	Gaussian	3	clean R2 below the gate	0.057
hERG K_i	BNN-β (var. head)	ChemBERTa	Gaussian	4	clean R2 below the gate	-0.022
hERG K_i	BNN-β (var. head)	ChemBERTa	Grouped, wider	0	clean R2 below the gate	-0.061
hERG K_i	BNN-β (var. head)	ChemBERTa	Grouped, wider	1	clean R2 below the gate	0.138
hERG K_i	BNN-β (var. head)	ChemBERTa	Grouped, wider	2	clean R2 below the gate	0.039
hERG K_i	BNN-β (var. head)	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.296
hERG K_i	BNN-β (var. head)	ChemBERTa	Grouped, wider	4	clean R2 below the gate	-0.038
hERG K_i	BNN-β (var. head)	ChemBERTa	Grouped, shifted	0	clean R2 below the gate	-0.060
hERG K_i	BNN-β (var. head)	ChemBERTa	Grouped, shifted	1	clean R2 below the gate	0.056
hERG K_i	BNN-β (var. head)	ChemBERTa	Grouped, shifted	2	clean R2 below the gate	0.026
hERG K_i	BNN-β (var. head)	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.039
hERG K_i	BNN-β (var. head)	MHG-GNN	Gaussian	0	clean R2 below the gate	-0.388
hERG K_i	BNN-β (var. head)	MHG-GNN	Gaussian	2	clean R2 below the gate	0.036

Dataset	Model	Representation	Condition	Fold	Reason	Clean R^2
hERG K_i	BNN-β (var. head)	MHG-GNN	Gaussian	3	clean R2 below the gate	-0.024
hERG K_i	BNN-β (var. head)	MHG-GNN	Gaussian	4	clean R2 below the gate	-0.147
hERG K_i	BNN-β (var. head)	MHG-GNN	Grouped, wider	2	clean R2 below the gate	-0.075
hERG K_i	BNN-β (var. head)	MHG-GNN	Grouped, shifted	2	clean R2 below the gate	0.024
hERG K_i	BNN-β (var. head)	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	-0.116
hERG K_i	BNN-β (var. head)	PDV	Gaussian	0	clean R2 below the gate	0.039
hERG K_i	BNN-β (var. head)	PDV	Gaussian	2	clean R2 below the gate	0.072
hERG K_i	BNN-β (var. head)	PDV	Gaussian	3	clean R2 below the gate	0.083
hERG K_i	BNN-β (var. head)	PDV	Gaussian	4	clean R2 below the gate	0.023
hERG K_i	BNN-β (var. head)	PDV	Grouped, wider	2	clean R2 below the gate	0.111
hERG K_i	BNN-β (var. head)	PDV	Grouped, wider	4	clean R2 below the gate	0.151
hERG K_i	BNN-β (var. head)	PDV	Grouped, shifted	0	clean R2 below the gate	0.107
hERG K_i	BNN-β (var. head)	PDV	Grouped, shifted	1	clean R2 below the gate	0.120
hERG K_i	BNN-β (var. head)	PDV	Grouped, shifted	2	clean R2 below the gate	0.190
hERG K_i	BNN-β (var. head)	PDV	Grouped, shifted	4	clean R2 below the gate	0.197
hERG K_i	BNN-β (var. head)	Sort & Slice	Gaussian	4	clean R2 below the gate	0.047
hERG K_i	BNN-β (var. head)	Sort & Slice	Grouped, wider	1	clean R2 below the gate	0.114
hERG K_i	BNN-β (var. head)	Sort & Slice	Grouped, wider	2	clean R2 below the gate	0.007
hERG K_i	BNN-β (var. head)	Sort & Slice	Grouped, shifted	0	clean R2 below the gate	0.062
hERG K_i	VBLL-β	ChemBERTa	Gaussian	4	clean R2 below the gate	0.282
hERG K_i	VBLL-β	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.287
hERG K_i	VBLL-β	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.277
hERG K_i	VBLL-β	MHG-GNN	Gaussian	4	clean R2 below the gate	0.258
hERG K_i	VBLL-β	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.260
hERG K_i	VBLL-β	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	0.295
hERG K_i	VBLL-β	PDV	Gaussian	4	clean R2 below the gate	0.264
hERG K_i	VBLL-β	PDV	Grouped, wider	4	clean R2 below the gate	0.265
hERG K_i	VBLL-β	PDV	Grouped, shifted	4	clean R2 below the gate	0.271
hERG K_i	VBLL-β (het.)	ChemBERTa	Gaussian	4	clean R2 below the gate	0.198
hERG K_i	VBLL-β (het.)	ChemBERTa	Grouped, wider	2	clean R2 below the gate	0.281
hERG K_i	VBLL-β (het.)	ChemBERTa	Grouped, wider	3	clean R2 below the gate	0.291
hERG K_i	VBLL-β (het.)	ChemBERTa	Grouped, wider	4	clean R2 below the gate	0.184
hERG K_i	VBLL-β (het.)	ChemBERTa	Grouped, shifted	3	clean R2 below the gate	0.295
hERG K_i	VBLL-β (het.)	ChemBERTa	Grouped, shifted	4	clean R2 below the gate	0.198
hERG K_i	VBLL-β (het.)	ChemBERTa	Student-t (ν=5)	0	clean R2 below the gate	0.286
hERG K_i	VBLL-β (het.)	ChemBERTa	Student-t (ν=5)	4	clean R2 below the gate	0.218
hERG K_i	VBLL-β (het.)	ChemBERTa	Outlier (10%)	2	clean R2 below the gate	0.286
hERG K_i	VBLL-β (het.)	ChemBERTa	Outlier (10%)	4	clean R2 below the gate	0.198
hERG K_i	VBLL-β (het.)	ChemBERTa	Laplace	2	clean R2 below the gate	0.286
hERG K_i	VBLL-β (het.)	ChemBERTa	Laplace	3	clean R2 below the gate	0.272
hERG K_i	VBLL-β (het.)	ChemBERTa	Laplace	4	clean R2 below the gate	0.191
hERG K_i	VBLL-β (het.)	MHG-GNN	Gaussian	1	clean R2 below the gate	0.297
hERG K_i	VBLL-β (het.)	MHG-GNN	Gaussian	2	clean R2 below the gate	0.260
hERG K_i	VBLL-β (het.)	MHG-GNN	Gaussian	4	clean R2 below the gate	0.220
hERG K_i	VBLL-β (het.)	MHG-GNN	Grouped, wider	2	clean R2 below the gate	0.272
hERG K_i	VBLL-β (het.)	MHG-GNN	Grouped, wider	4	clean R2 below the gate	0.258
hERG K_i	VBLL-β (het.)	MHG-GNN	Grouped, shifted	4	clean R2 below the gate	0.254
hERG K_i	VBLL-β (het.)	PDV	Gaussian	2	clean R2 below the gate	0.234
hERG K_i	VBLL-β (het.)	PDV	Gaussian	3	clean R2 below the gate	0.266
hERG K_i	VBLL-β (het.)	PDV	Gaussian	4	clean R2 below the gate	0.176
hERG K_i	VBLL-β (het.)	PDV	Grouped, wider	3	clean R2 below the gate	0.299
hERG K_i	VBLL-β (het.)	PDV	Grouped, wider	4	clean R2 below the gate	0.183
hERG K_i	VBLL-β (het.)	PDV	Grouped, shifted	2	clean R2 below the gate	0.264
hERG K_i	VBLL-β (het.)	PDV	Grouped, shifted	3	clean R2 below the gate	0.049
hERG K_i	VBLL-β (het.)	PDV	Grouped, shifted	4	clean R2 below the gate	0.226
hERG K_i	VBLL-β (het.)	PDV	Student-t (ν=5)	2	clean R2 below the gate	0.270
hERG K_i	VBLL-β (het.)	PDV	Student-t (ν=5)	3	clean R2 below the gate	-0.087
hERG K_i	VBLL-β (het.)	PDV	Student-t (ν=5)	4	clean R2 below the gate	0.220
hERG K_i	VBLL-β (het.)	PDV	Outlier (10%)	2	clean R2 below the gate	0.285
hERG K_i	VBLL-β (het.)	PDV	Outlier (10%)	3	clean R2 below the gate	0.274
hERG K_i	VBLL-β (het.)	PDV	Outlier (10%)	4	clean R2 below the gate	0.224
hERG K_i	VBLL-β (het.)	PDV	Laplace	2	clean R2 below the gate	0.293
hERG K_i	VBLL-β (het.)	PDV	Laplace	3	clean R2 below the gate	0.279
hERG K_i	VBLL-β (het.)	PDV	Laplace	4	clean R2 below the gate	0.179

Additional file 3: Candidate model families

For each pair of models considered for grouping, on how many of the six representations the two differ by less than their standard deviation across replicates, for clean R^2 and for $\mathrm{AUC}_{\mathrm{norm}}$ under each noise condition run on every representation.

Table S.5: Candidate model families: how often two models differ by less than their split-to-split variation. Each row is one pair of base models considered for a family. Each cell counts the representations, of six, on which the two differ by less than their split-to-split standard deviation, followed by the range of the absolute difference. Grouped says whether the pair shares a family in the ANOVA. Outcomes are clean R^2 and $\mathrm{AUC}_{\mathrm{norm}}$ under the three noise conditions every representation ran, on QM9. The difference is the mean over the ten splits of one member minus the other, paired on the split; the split-to-split standard deviation is that of the two members' values over the splits.

Model A	Model B	Grouped	Clean R^2	Gaussian	Grouped, shifted	Grouped, wider
RF	QRF	yes	6 of 6 (0.000–0.008)	2 of 6 (0.005–0.010)	6 of 6 (0.011–0.016)	2 of 6 (0.001–0.009)
XGBoost	LightGBM	yes	5 of 6 (0.001–0.022)	6 of 6 (0.000–0.006)	6 of 6 (0.001–0.016)	6 of 6 (0.000–0.004)
XGBoost	NGBoost	no	0 of 6 (0.044–0.125)	0 of 6 (0.015–0.043)	3 of 6 (0.007–0.047)	0 of 6 (0.008–0.028)
LightGBM	NGBoost	no	0 of 6 (0.045–0.138)	0 of 6 (0.021–0.040)	2 of 6 (0.018–0.041)	0 of 6 (0.011–0.028)
NN-α	BNN-α	yes	5 of 6 (0.001–0.023)	6 of 6 (0.004–0.016)	6 of 6 (0.002–0.026)	5 of 6 (0.002–0.025)
NN-α	VBLL-α	no	0 of 6 (0.024–0.073)	2 of 6 (0.008–0.023)	0 of 6 (0.026–0.079)	2 of 6 (0.008–0.019)
BNN-α	VBLL-α	no	0 of 6 (0.042–0.066)	2 of 6 (0.004–0.039)	2 of 6 (0.006–0.059)	2 of 6 (0.006–0.043)
NN-β	BNN-β	yes	5 of 6 (0.000–0.054)	5 of 6 (0.003–0.021)	5 of 6 (0.002–0.033)	4 of 6 (0.001–0.017)
NN-β	VBLL-β	no	0 of 6 (0.044–0.089)	2 of 6 (0.004–0.048)	0 of 6 (0.045–0.067)	3 of 6 (0.005–0.040)
BNN-β	VBLL-β	no	1 of 6 (0.035–0.076)	4 of 6 (0.003–0.056)	1 of 6 (0.017–0.074)	5 of 6 (0.003–0.058)
VBLL-α	VBLL-β	yes	5 of 6 (0.001–0.037)	5 of 6 (0.003–0.017)	6 of 6 (0.001–0.010)	4 of 6 (0.001–0.022)
NN-α	NN-β	no	4 of 6 (0.003–0.045)	4 of 6 (0.004–0.035)	5 of 6 (0.007–0.042)	4 of 6 (0.002–0.031)
BNN-α	BNN-β	no	4 of 6 (0.005–0.063)	4 of 6 (0.005–0.032)	6 of 6 (0.002–0.031)	4 of 6 (0.004–0.025)

Additional file 4: Tukey's honestly significant difference test within each model family

For each pair of models in a family, on how many representations the two differ at adjusted $p < 0.05$, with the size of the difference.

Table S.6: Tukey HSD between the members of each model family. One test per representation and outcome, after removing each split's mean. Each cell counts the representations, of six, on which the pair differs at adjusted $p < 0.05$, followed by the range of the absolute difference. With ten splits the test detects differences smaller than the split-to-split variation, so the size of each difference is given beside it.

Family	Model A	Model B	Clean R^2	Gaussian	Grouped, shifted	Grouped, wider
Forest	QRF	RF	3 of 6 (0.000–0.008)	6 of 6 (0.005–0.010)	6 of 6 (0.011–0.016)	5 of 6 (0.001–0.009)
Boosting	LightGBM	XGBoost	4 of 6 (0.001–0.022)	3 of 6 (0.000–0.006)	4 of 6 (0.001–0.017)	1 of 6 (0.000–0.004)
NN-α	NN-α	BNN-α	5 of 6 (0.001–0.023)	3 of 6 (0.004–0.016)	4 of 6 (0.002–0.026)	2 of 6 (0.002–0.025)

Family	Model A	Model B	Clean R^2	Gaussian	Grouped, shifted	Grouped, wider
NN-β	NN-β	BNN-β	1 of 6 (0.000–0.054)	2 of 6 (0.003–0.021)	2 of 6 (0.002–0.033)	2 of 6 (0.001–0.018)
VBLL	VBLL-α	VBLL-β	5 of 6 (0.001–0.037)	4 of 6 (0.003–0.017)	2 of 6 (0.001–0.010)	4 of 6 (0.001–0.022)

Additional file 5: Candidate representation groups

The same comparison as Additional file 3, for pairs of representations over the models.

Table S.7: Candidate representation groups: how often two representations differ by less than their split-to-split variation. Each row is one pair of representations considered for a group. Each cell counts the base models on which the two differ by less than their split-to-split standard deviation, followed by the range of the absolute difference. No pair was grouped. Outcomes are clean R^2 and $\mathrm{AUC}_{\mathrm{norm}}$ under the three noise conditions every representation ran, on QM9. The difference is the mean over the ten splits of one member minus the other, paired on the split; the split-to-split standard deviation is that of the two members' values over the splits.

Representation A	Representation B	Clean R^2	Gaussian	Grouped, shifted	Grouped, wider
ECFP4	Sort & Slice	0 of 13 (0.035–0.105)	5 of 13 (0.001–0.041)	12 of 13 (0.005–0.025)	5 of 13 (0.001–0.047)

Additional file 6: Intervals for the shares of variance

Every share in the variance decomposition tables, with the lower and upper ends of its 95% interval from 1,000 bootstrap resamples of whole replicates, on all four datasets.

Table S.8: Intervals for the shares of variance in the decomposition tables. One row is one term of one decomposition: a dataset, a noise condition and an outcome. The share is the one printed in Chapter 3; the lower and upper ends are its 95% interval from 1,000 bootstrap resamples of whole replicates. QM9 has 10 replicates; on each assay dataset a replicate is one of five scaffold folds, so its interval rests on five. Predictive performance under noise is R^2 at each dataset's reporting noise level: 1.0 on QM9, LogD and hERG K_i, and 0.75 on Caco-2.

Dataset	Condition	Outcome	Term	Share (%)	Lower (%)	Upper (%)
QM9	None (clean labels)	Predictive performance (R^2 on clean labels)	Replicate	10.6	2.0	16.9
QM9	None (clean labels)	Predictive performance (R^2 on clean labels)	Model family	35.4	30.5	42.1
QM9	None (clean labels)	Predictive performance (R^2 on clean labels)	Model within family	0.6	0.5	0.8
QM9	None (clean labels)	Predictive performance (R^2 on clean labels)	Representation	31.1	28.6	36.1
QM9	None (clean labels)	Predictive performance (R^2 on clean labels)	Interaction	12.1	9.7	16.8
QM9	None (clean labels)	Predictive performance (R^2 on clean labels)	Residual	10.2	5.7	11.9
QM9	Gaussian	Robustness (AUC_{norm})	Replicate	1.7	0.5	2.8
QM9	Gaussian	Robustness (AUC_{norm})	Model family	47.2	41.3	54.1
QM9	Gaussian	Robustness (AUC_{norm})	Model within family	0.6	0.4	1.3
QM9	Gaussian	Robustness (AUC_{norm})	Representation	8.1	6.0	11.2
QM9	Gaussian	Robustness (AUC_{norm})	Interaction	19.9	17.6	27.8
QM9	Gaussian	Robustness (AUC_{norm})	Residual	22.6	14.3	24.8
QM9	Grouped, wider	Robustness (AUC_{norm})	Replicate	1.7	0.6	2.6
QM9	Grouped, wider	Robustness (AUC_{norm})	Model family	46.3	40.0	54.0
QM9	Grouped, wider	Robustness (AUC_{norm})	Model within family	0.3	0.1	0.7
QM9	Grouped, wider	Robustness (AUC_{norm})	Representation	7.9	6.4	10.2
QM9	Grouped, wider	Robustness (AUC_{norm})	Interaction	18.5	16.2	27.9
QM9	Grouped, wider	Robustness (AUC_{norm})	Residual	25.3	16.4	27.7
QM9	Grouped, shifted	Robustness (AUC_{norm})	Replicate	28.4	2.1	38.1
QM9	Grouped, shifted	Robustness (AUC_{norm})	Model family	32.7	25.2	49.5
QM9	Grouped, shifted	Robustness (AUC_{norm})	Model within family	1.0	0.7	1.9
QM9	Grouped, shifted	Robustness (AUC_{norm})	Representation	7.0	5.0	12.1
QM9	Grouped, shifted	Robustness (AUC_{norm})	Interaction	7.2	6.8	11.8
QM9	Grouped, shifted	Robustness (AUC_{norm})	Residual	23.7	17.0	27.5
QM9	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Replicate	13.4	1.8	19.7
QM9	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Model family	26.6	22.1	33.7
QM9	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Model within family	0.8	0.6	1.3
QM9	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Representation	29.0	27.3	33.8
QM9	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Interaction	14.2	12.2	18.7
QM9	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Residual	16.0	11.7	16.6
QM9	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Replicate	12.7	1.5	19.4
QM9	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Model family	33.0	28.8	40.4

Dataset	Condition	Outcome	Term	Share (%)	Lower (%)	Upper (%)
QM9	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Model within family	0.4	0.2	1.0
QM9	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Representation	25.7	22.3	31.0
QM9	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Interaction	13.1	12.0	17.0
QM9	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Residual	15.1	10.3	16.5
QM9	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Replicate	51.0	6.5	69.0
QM9	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Model family	8.4	2.4	26.4
QM9	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Model within family	0.4	0.2	1.8
QM9	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Representation	10.3	5.0	27.0
QM9	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Interaction	6.2	4.7	15.8
QM9	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Residual	23.7	16.3	32.9
LogD	None (clean labels)	Predictive performance (R^2 on clean labels)	Replicate	3.3	0.2	4.7
LogD	None (clean labels)	Predictive performance (R^2 on clean labels)	Model family	60.2	58.9	63.2
LogD	None (clean labels)	Predictive performance (R^2 on clean labels)	Model within family	6.7	5.7	8.1
LogD	None (clean labels)	Predictive performance (R^2 on clean labels)	Representation	16.8	14.3	20.1
LogD	None (clean labels)	Predictive performance (R^2 on clean labels)	Interaction	9.5	9.1	11.2
LogD	None (clean labels)	Predictive performance (R^2 on clean labels)	Residual	3.6	1.4	4.0
LogD	Gaussian	Robustness (AUC_{norm})	Replicate	1.3	0.1	2.4
LogD	Gaussian	Robustness (AUC_{norm})	Model family	58.0	49.7	67.1
LogD	Gaussian	Robustness (AUC_{norm})	Model within family	5.2	3.9	7.5
LogD	Gaussian	Robustness (AUC_{norm})	Representation	7.1	5.7	9.9
LogD	Gaussian	Robustness (AUC_{norm})	Interaction	12.2	10.9	19.2
LogD	Gaussian	Robustness (AUC_{norm})	Residual	16.3	5.6	18.5
LogD	Grouped, wider	Robustness (AUC_{norm})	Replicate	2.5	0.1	4.2
LogD	Grouped, wider	Robustness (AUC_{norm})	Model family	56.1	47.3	65.6
LogD	Grouped, wider	Robustness (AUC_{norm})	Model within family	3.6	2.0	6.3
LogD	Grouped, wider	Robustness (AUC_{norm})	Representation	7.5	6.9	9.6
LogD	Grouped, wider	Robustness (AUC_{norm})	Interaction	13.4	13.4	22.9
LogD	Grouped, wider	Robustness (AUC_{norm})	Residual	16.8	6.5	17.9
LogD	Grouped, shifted	Robustness (AUC_{norm})	Replicate	13.9	0.6	26.3
LogD	Grouped, shifted	Robustness (AUC_{norm})	Model family	26.2	22.2	35.7
LogD	Grouped, shifted	Robustness (AUC_{norm})	Model within family	7.5	4.7	14.5
LogD	Grouped, shifted	Robustness (AUC_{norm})	Representation	11.2	10.2	15.4
LogD	Grouped, shifted	Robustness (AUC_{norm})	Interaction	13.3	10.3	26.8
LogD	Grouped, shifted	Robustness (AUC_{norm})	Residual	27.9	13.7	30.2
LogD	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Replicate	4.5	0.6	7.4
LogD	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Model family	59.6	56.2	66.0
LogD	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Model within family	5.6	3.4	8.8
LogD	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Representation	14.8	11.4	19.3
LogD	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Interaction	9.3	9.1	12.0
LogD	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Residual	6.3	2.6	6.9
LogD	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Replicate	4.6	0.4	6.7
LogD	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Model family	62.4	59.9	69.6
LogD	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Model within family	4.1	3.1	5.9

Dataset	Condition	Outcome	Term	Share (%)	Lower (%)	Upper (%)
LogD	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Representation	15.3	10.9	20.2
LogD	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Interaction	7.7	7.4	9.9
LogD	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Residual	5.9	2.8	5.9
LogD	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Replicate	12.1	1.0	15.9
LogD	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Model family	34.7	30.8	48.0
LogD	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Model within family	4.7	3.4	7.3
LogD	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Representation	25.0	19.3	33.9
LogD	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Interaction	11.2	10.2	16.7
LogD	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Residual	12.4	5.2	13.1
Caco-2	None (clean labels)	Predictive performance (R^2 on clean labels)	Replicate	38.7	0.6	47.6
Caco-2	None (clean labels)	Predictive performance (R^2 on clean labels)	Model family	22.8	19.3	41.6
Caco-2	None (clean labels)	Predictive performance (R^2 on clean labels)	Model within family	13.4	10.8	22.0
Caco-2	None (clean labels)	Predictive performance (R^2 on clean labels)	Representation	9.9	8.4	16.4
Caco-2	None (clean labels)	Predictive performance (R^2 on clean labels)	Interaction	8.9	7.9	17.2
Caco-2	None (clean labels)	Predictive performance (R^2 on clean labels)	Residual	6.2	2.7	8.9
Caco-2	Gaussian	Robustness (AUC_{norm})	Replicate	18.6	0.1	25.7
Caco-2	Gaussian	Robustness (AUC_{norm})	Model family	43.2	37.5	60.6
Caco-2	Gaussian	Robustness (AUC_{norm})	Model within family	4.8	3.3	9.6
Caco-2	Gaussian	Robustness (AUC_{norm})	Representation	3.1	2.3	5.0
Caco-2	Gaussian	Robustness (AUC_{norm})	Interaction	9.4	9.2	20.9
Caco-2	Gaussian	Robustness (AUC_{norm})	Residual	20.8	7.3	23.6
Caco-2	Grouped, wider	Robustness (AUC_{norm})	Replicate	17.8	1.4	27.7
Caco-2	Grouped, wider	Robustness (AUC_{norm})	Model family	46.7	39.2	66.9
Caco-2	Grouped, wider	Robustness (AUC_{norm})	Model within family	1.8	1.3	5.7
Caco-2	Grouped, wider	Robustness (AUC_{norm})	Representation	3.2	1.2	7.2
Caco-2	Grouped, wider	Robustness (AUC_{norm})	Interaction	8.8	8.0	22.7
Caco-2	Grouped, wider	Robustness (AUC_{norm})	Residual	21.7	9.0	22.8
Caco-2	Grouped, shifted	Robustness (AUC_{norm})	Replicate	39.7	3.3	48.9
Caco-2	Grouped, shifted	Robustness (AUC_{norm})	Model family	29.1	23.6	49.7
Caco-2	Grouped, shifted	Robustness (AUC_{norm})	Model within family	3.6	1.7	12.4
Caco-2	Grouped, shifted	Robustness (AUC_{norm})	Representation	4.4	3.5	9.6
Caco-2	Grouped, shifted	Robustness (AUC_{norm})	Interaction	5.2	4.7	17.3
Caco-2	Grouped, shifted	Robustness (AUC_{norm})	Residual	18.1	8.5	20.0
Caco-2	Gaussian	Predictive performance (R^2 at a noise level of 0.75)	Replicate	56.0	2.2	66.2
Caco-2	Gaussian	Predictive performance (R^2 at a noise level of 0.75)	Model family	11.4	8.2	31.9
Caco-2	Gaussian	Predictive performance (R^2 at a noise level of 0.75)	Model within family	5.8	4.0	13.8
Caco-2	Gaussian	Predictive performance (R^2 at a noise level of 0.75)	Representation	9.7	6.9	23.3
Caco-2	Gaussian	Predictive performance (R^2 at a noise level of 0.75)	Interaction	6.6	5.4	18.8
Caco-2	Gaussian	Predictive performance (R^2 at a noise level of 0.75)	Residual	10.5	5.8	20.2
Caco-2	Grouped, wider	Predictive performance (R^2 at a noise level of 0.75)	Replicate	53.4	2.7	65.6
Caco-2	Grouped, wider	Predictive performance (R^2 at a noise level of 0.75)	Model family	14.0	10.6	33.8
Caco-2	Grouped, wider	Predictive performance (R^2 at a noise level of 0.75)	Model within family	6.0	4.2	12.9
Caco-2	Grouped, wider	Predictive performance (R^2 at a noise level of 0.75)	Representation	8.7	6.1	18.2

Dataset	Condition	Outcome	Term	Share (%)	Lower (%)	Upper (%)
Caco-2	Grouped, wider	Predictive performance (R^2 at a noise level of 0.75)	Interaction	8.2	6.7	22.3
Caco-2	Grouped, wider	Predictive performance (R^2 at a noise level of 0.75)	Residual	9.7	5.2	15.5
Caco-2	Grouped, shifted	Predictive performance (R^2 at a noise level of 0.75)	Replicate	67.9	11.7	78.6
Caco-2	Grouped, shifted	Predictive performance (R^2 at a noise level of 0.75)	Model family	4.4	3.1	18.6
Caco-2	Grouped, shifted	Predictive performance (R^2 at a noise level of 0.75)	Model within family	2.3	1.4	10.3
Caco-2	Grouped, shifted	Predictive performance (R^2 at a noise level of 0.75)	Representation	7.3	5.1	20.2
Caco-2	Grouped, shifted	Predictive performance (R^2 at a noise level of 0.75)	Interaction	3.6	3.0	17.7
Caco-2	Grouped, shifted	Predictive performance (R^2 at a noise level of 0.75)	Residual	14.4	6.7	32.3
hERG K_i	None (clean labels)	Predictive performance (R^2 on clean labels)	Replicate	28.1	2.5	35.6
hERG K_i	None (clean labels)	Predictive performance (R^2 on clean labels)	Model family	18.4	15.9	29.0
hERG K_i	None (clean labels)	Predictive performance (R^2 on clean labels)	Model within family	15.1	9.5	27.6
hERG K_i	None (clean labels)	Predictive performance (R^2 on clean labels)	Representation	6.7	4.2	14.3
hERG K_i	None (clean labels)	Predictive performance (R^2 on clean labels)	Interaction	14.3	13.2	23.6
hERG K_i	None (clean labels)	Predictive performance (R^2 on clean labels)	Residual	17.5	7.4	21.2
hERG K_i	Gaussian	Robustness (AUC_{norm})	Replicate	5.0	0.7	10.0
hERG K_i	Gaussian	Robustness (AUC_{norm})	Model family	47.9	45.8	55.6
hERG K_i	Gaussian	Robustness (AUC_{norm})	Model within family	5.3	3.9	8.5
hERG K_i	Gaussian	Robustness (AUC_{norm})	Representation	0.8	0.4	3.6
hERG K_i	Gaussian	Robustness (AUC_{norm})	Interaction	13.0	13.0	23.1
hERG K_i	Gaussian	Robustness (AUC_{norm})	Residual	28.0	8.7	31.2
hERG K_i	Grouped, wider	Robustness (AUC_{norm})	Replicate	13.2	0.2	16.9
hERG K_i	Grouped, wider	Robustness (AUC_{norm})	Model family	40.5	37.1	55.2
hERG K_i	Grouped, wider	Robustness (AUC_{norm})	Model within family	4.4	3.0	7.7
hERG K_i	Grouped, wider	Robustness (AUC_{norm})	Representation	2.9	2.3	4.9
hERG K_i	Grouped, wider	Robustness (AUC_{norm})	Interaction	6.3	6.3	18.2
hERG K_i	Grouped, wider	Robustness (AUC_{norm})	Residual	32.7	12.2	34.9
hERG K_i	Grouped, shifted	Robustness (AUC_{norm})	Replicate	17.3	2.1	29.3
hERG K_i	Grouped, shifted	Robustness (AUC_{norm})	Model family	34.6	29.6	51.1
hERG K_i	Grouped, shifted	Robustness (AUC_{norm})	Model within family	4.0	2.3	8.7
hERG K_i	Grouped, shifted	Robustness (AUC_{norm})	Representation	0.9	0.4	3.8
hERG K_i	Grouped, shifted	Robustness (AUC_{norm})	Interaction	14.2	10.9	31.4
hERG K_i	Grouped, shifted	Robustness (AUC_{norm})	Residual	29.0	9.4	31.4
hERG K_i	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Replicate	39.4	6.0	53.8
hERG K_i	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Model family	22.5	14.8	45.6
hERG K_i	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Model within family	2.1	0.9	6.6
hERG K_i	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Representation	5.6	3.8	14.0
hERG K_i	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Interaction	7.0	6.6	20.9
hERG K_i	Gaussian	Predictive performance (R^2 at a noise level of 1.0)	Residual	23.5	10.1	29.6
hERG K_i	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Replicate	29.4	3.0	43.3
hERG K_i	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Model family	29.3	25.0	39.5
hERG K_i	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Model within family	2.5	0.6	7.7
hERG K_i	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Representation	5.6	2.9	12.2
hERG K_i	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Interaction	11.6	10.1	26.2

Dataset	Condition	Outcome	Term	Share (%)	Lower (%)	Upper (%)
hERG K_i	Grouped, wider	Predictive performance (R^2 at a noise level of 1.0)	Residual	21.6	10.4	22.5
hERG K_i	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Replicate	32.6	5.4	47.3
hERG K_i	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Model family	15.6	12.8	29.0
hERG K_i	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Model within family	2.5	1.6	7.5
hERG K_i	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Representation	1.9	0.7	8.5
hERG K_i	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Interaction	12.5	10.1	29.8
hERG K_i	Grouped, shifted	Predictive performance (R^2 at a noise level of 1.0)	Residual	35.0	17.4	39.4

Additional file 7: Robustness grid under grouped-wider noise

AUC_{norm} of every base model on every representation under grouped-wider noise on QM9, the condition Figure 3.6 does not show.

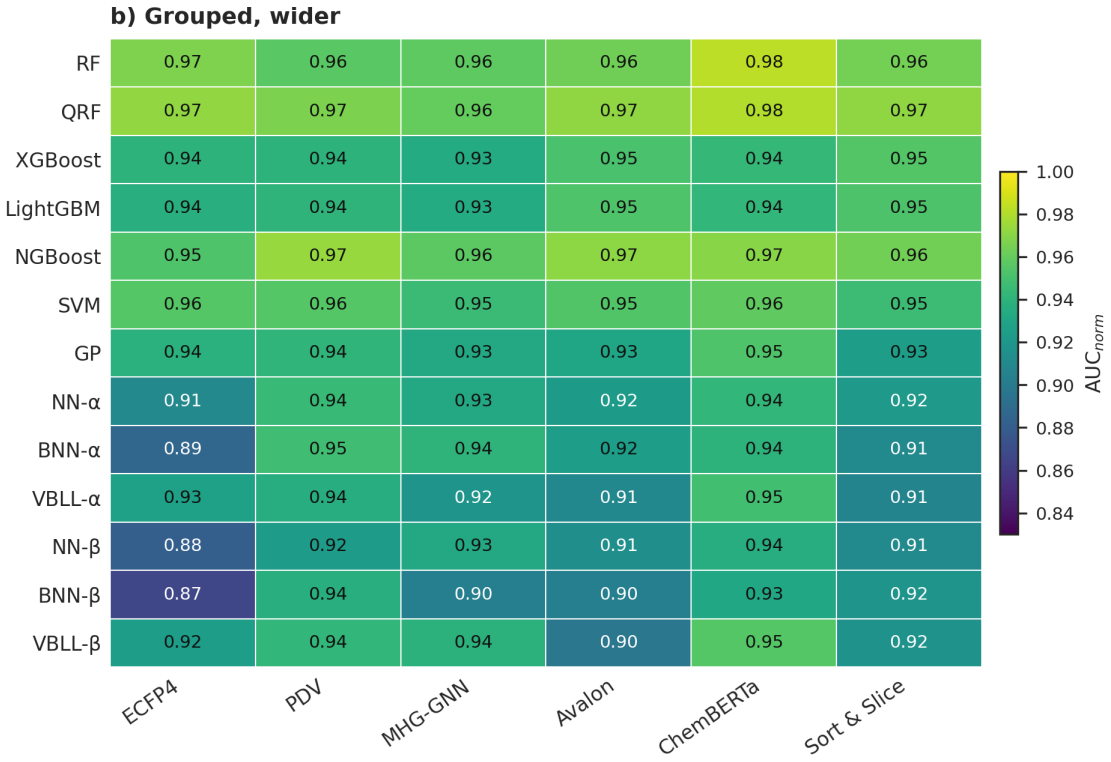


Figure S.1: Robustness of every base model on every representation under grouped-wider noise, on the QM9 HOMO–LUMO gap. Rows are the 13 base models, ordered by family, and columns the six representations. One cell is one model on one representation, the median AUC_{norm} over 10 replicates, with its value printed on it. Color runs on the same fixed range as Figure 3.6, which shows Gaussian and grouped-shifted noise.

Additional file 8: Robustness of the variant models on the three assay datasets

Clean R^2 and $\mathrm{AUC}_{\mathrm{norm}}$ of the six variant models on LogD, Caco-2 and hERG K_i, on ECFP4, beside Figure 3.11.

Table S.9: Robustness of the six variant models on the three assay datasets, on ECFP4. One row is one variant model on one dataset. The first column is clean R^2; the others are $\mathrm{AUC}_{\mathrm{norm}}$ under each noise condition run on every model. Values are medians over five scaffold folds.

Dataset	Model	Clean R^2	Gaussian	Grouped, wider	Grouped, shifted
LogD	RF (300 trees)	0.643	0.947	0.958	0.897
LogD	GP (het.)	0.804	0.896	0.893	0.798
LogD	BNN-α (var. head)	0.748	0.881	0.852	0.773
LogD	VBLL-α (het.)	0.556	0.693	0.731	0.768
LogD	BNN-β (var. head)	0.752	0.884	0.867	0.797
LogD	VBLL-β (het.)	0.659	0.796	0.864	0.837
Caco-2	RF (300 trees)	0.412	0.876	0.878	0.774
Caco-2	GP (het.)	0.517	0.815	0.834	0.706
Caco-2	BNN-α (var. head)	0.394	0.787	0.769	0.721
Caco-2	VBLL-α (het.)	0.329	0.653	0.733	0.781
Caco-2	BNN-β (var. head)	0.454	0.739	0.739	0.766
Caco-2	VBLL-β (het.)	0.414	0.707	0.739	0.806
hERG K_i	RF (300 trees)	0.553	0.910	0.917	0.800
hERG K_i	GP (het.)	0.578	0.843	0.864	0.775
hERG K_i	BNN-α (var. head)	0.491	0.809	0.800	0.679
hERG K_i	VBLL-α (het.)	0.420	0.730	0.845	0.749
hERG K_i	BNN-β (var. head)	0.539	0.812	0.753	0.728
hERG K_i	VBLL-β (het.)	0.410	0.836	0.834	0.811

Additional file 9: Base models against their probabilistic counterparts on the three assay datasets

The change in $\mathrm{AUC}_{\mathrm{norm}}$ and in clean R^2, counterpart minus base, for each pair, noise condition and representation, on LogD, Caco-2 and hERG K_i. These are the assay-dataset versions of Table **??**.

Table S.10: LogD: what replacing a model with its own probabilistic counterpart changes. Two rows per pair and noise condition, one column per representation: the change in $\mathrm{AUC}_{\mathrm{norm}}$, then the change in clean R^2, both counterpart minus base, the median of the paired differences over five scaffold folds. † marks an $\mathrm{AUC}_{\mathrm{norm}}$ change with the same sign in all five. "Settings" says whether the two models share one setting.

Comparison	Settings	Condition	Change	ECFP4	PDV	MHG-GNN	Avalon	ChemBERTa	Sort & Slice
RF (300 trees) → QRF	shared	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	-0.017	-0.020†	-0.021†	-0.021†	-0.019†	-0.021†
RF (300 trees) → QRF	shared	Gaussian	Clean R^2	-0.005	+0.009	+0.008	-0.001	+0.014	-0.004
RF (300 trees) → QRF	shared	Grouped, shifted	$\mathrm{AUC}_{\mathrm{norm}}$	-0.037†	-0.027†	-0.032†	-0.032†	-0.023†	-0.045
RF (300 trees) → QRF	shared	Grouped, shifted	Clean R^2	-0.005	+0.009	+0.008	-0.001	+0.014	-0.004
RF (300 trees) → QRF	shared	Grouped, wider	$\mathrm{AUC}_{\mathrm{norm}}$	+0.005†	-0.004†	-0.017	-0.003	-0.015†	-0.010
RF (300 trees) → QRF	shared	Grouped, wider	Clean R^2	-0.005	+0.009	+0.008	-0.001	+0.014	-0.004
RF → QRF	differ	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	-0.008†	-0.015†	-0.022†	-0.010	-0.005	-0.017†
RF → QRF	differ	Gaussian	Clean R^2	-0.005	+0.011	+0.012	+0.002	+0.020	-0.004
RF → QRF	differ	Grouped, shifted	$\mathrm{AUC}_{\mathrm{norm}}$	-0.034†	-0.015	-0.028†	-0.018†	-0.014	-0.033†
RF → QRF	differ	Grouped, shifted	Clean R^2	-0.005	+0.011	+0.012	+0.002	+0.020	-0.004
RF → QRF	differ	Grouped, wider	$\mathrm{AUC}_{\mathrm{norm}}$	+0.003	-0.001	-0.006	+0.001	+0.003	-0.007
RF → QRF	differ	Grouped, wider	Clean R^2	-0.005	+0.011	+0.012	+0.002	+0.020	-0.004
NN-α → BNN-α	shared	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	+0.037†	-0.023	-0.058	+0.003	-0.028	+0.040†
NN-α → BNN-α	shared	Gaussian	Clean R^2	-0.018	-0.094	-0.157	-0.084	-0.056	-0.060
NN-α → BNN-α	shared	Grouped, shifted	$\mathrm{AUC}_{\mathrm{norm}}$	+0.054†	+0.112	-0.087	+0.138†	+0.011	+0.029†
NN-α → BNN-α	shared	Grouped, shifted	Clean R^2	-0.028	-0.128	-0.049	-0.139	-0.045	-0.049
NN-α → BNN-α	shared	Grouped, wider	$\mathrm{AUC}_{\mathrm{norm}}$	+0.027†	-0.009	-0.075	+0.067†	-0.010	+0.038†
NN-α → BNN-α	shared	Grouped, wider	Clean R^2	-0.026	-0.062	-0.053	-0.144	-0.111	-0.053
BNN-α → VBLL-α	differ	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	-0.159†	-0.117	-0.075	-0.334†	-0.208†	-0.106†
BNN-α → VBLL-α	differ	Gaussian	Clean R^2	-0.234	-0.053	-0.136	-0.258	-0.270	-0.170
BNN-α → VBLL-α	differ	Grouped, shifted	$\mathrm{AUC}_{\mathrm{norm}}$	-0.112†	-0.115	+0.030	-0.183	-0.114	-0.061
BNN-α → VBLL-α	differ	Grouped, shifted	Clean R^2	-0.210	-0.012	-0.221	-0.261	-0.290	-0.135
BNN-α → VBLL-α	differ	Grouped, wider	$\mathrm{AUC}_{\mathrm{norm}}$	-0.124†	-0.041	-0.094†	-0.357†	-0.200†	-0.141†
BNN-α → VBLL-α	differ	Grouped, wider	Clean R^2	-0.215	-0.125	-0.200	-0.227	-0.235	-0.114
BNN-α → BNN-α (var. head)	differ	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	-0.007	-0.013	-0.251	+0.004	-0.060	-0.053
BNN-α → BNN-α (var. head)	differ	Gaussian	Clean R^2	-0.018	-0.027	-0.036	-0.013	-0.038	+0.016
BNN-α → BNN-α (var. head)	differ	Grouped, shifted	$\mathrm{AUC}_{\mathrm{norm}}$	-0.060†	-0.119	-0.047	-0.133†	-0.085†	-0.027
BNN-α → BNN-α (var. head)	differ	Grouped, shifted	Clean R^2	+0.005	+0.085	-0.123	+0.030	-0.064	+0.016
BNN-α → BNN-α (var. head)	differ	Grouped, wider	$\mathrm{AUC}_{\mathrm{norm}}$	-0.027	-0.025	-0.000	-0.105	-0.122†	-0.050†
BNN-α → BNN-α (var. head)	differ	Grouped, wider	Clean R^2	-0.000	+0.005	-0.175	-0.026	-0.016	+0.030
NN-β → BNN-β	shared	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	+0.035†	+0.074†	+0.009	+0.022	+0.021	+0.002
NN-β → BNN-β	shared	Gaussian	Clean R^2	-0.030	-0.103	-0.091	-0.093	-0.062	-0.016

Comparison	Settings	Condition	Change	ECFP4	PDV	MHG-GNN	Avalon	ChemBERTa	Sort & Slice
NN-β → BNN-β	shared	Grouped, shifted	AUC_{norm}	+0.043†	+0.026	+0.058†	+0.036	+0.037	+0.046†
NN-β → BNN-β	shared	Grouped, shifted	Clean R^2	-0.026	-0.036	-0.139	-0.061	-0.053	-0.051
NN-β → BNN-β	shared	Grouped, wider	AUC_{norm}	+0.039†	-0.009	-0.047	+0.013	-0.004	+0.040
NN-β → BNN-β	shared	Grouped, wider	Clean R^2	-0.024	-0.028	-0.040	-0.060	-0.034	-0.046
BNN-β → VBLL-β	differ	Gaussian	AUC_{norm}	-0.098†	-0.081†	-0.118†	-0.095†	-0.153†	-0.046
BNN-β → VBLL-β	differ	Gaussian	Clean R^2	-0.144	-0.029	-0.111	-0.130	-0.167	-0.095
BNN-β → VBLL-β	differ	Grouped, shifted	AUC_{norm}	-0.042	-0.010	-0.061†	-0.051	-0.053	+0.015
BNN-β → VBLL-β	differ	Grouped, shifted	Clean R^2	-0.118	-0.062	-0.103	-0.109	-0.166	-0.079
BNN-β → VBLL-β	differ	Grouped, wider	AUC_{norm}	-0.077†	-0.045†	-0.079	-0.096	-0.086†	-0.060†
BNN-β → VBLL-β	differ	Grouped, wider	Clean R^2	-0.133	-0.064	-0.145	-0.132	-0.178	-0.044
BNN-β → BNN-β (var. head)	differ	Gaussian	AUC_{norm}	-0.008	+0.013	-0.041	+0.018	-0.169	+0.011
BNN-β → BNN-β (var. head)	differ	Gaussian	Clean R^2	-0.013	-0.006	-0.045	-0.014	-0.042	-0.031
BNN-β → BNN-β (var. head)	differ	Grouped, shifted	AUC_{norm}	-0.038†	-0.052	-0.153	+0.063	-0.035	-0.043
BNN-β → BNN-β (var. head)	differ	Grouped, shifted	Clean R^2	-0.011	+0.012	+0.016	-0.060	-0.048	+0.007
BNN-β → BNN-β (var. head)	differ	Grouped, wider	AUC_{norm}	-0.009	-0.025	-0.176†	-0.008	-0.102	-0.007
BNN-β → BNN-β (var. head)	differ	Grouped, wider	Clean R^2	-0.014	-0.000	-0.085	-0.039	-0.000	+0.007
GP → GP (het.)	shared	Gaussian	AUC_{norm}	+0.006	-0.001	-0.006	+0.000	-0.004	-0.005
GP → GP (het.)	shared	Gaussian	Clean R^2	+0.000	+0.000	+0.000	+0.000	-0.000	-0.000
GP → GP (het.)	shared	Grouped, shifted	AUC_{norm}	-0.003	+0.010	+0.022	+0.020†	+0.024†	-0.003
GP → GP (het.)	shared	Grouped, shifted	Clean R^2	+0.000	-0.000	+0.000	+0.000	-0.001	-0.000
GP → GP (het.)	shared	Grouped, wider	AUC_{norm}	-0.009	-0.002	+0.002	-0.001	-0.002	-0.006
GP → GP (het.)	shared	Grouped, wider	Clean R^2	+0.000	-0.000	+0.000	+0.000	-0.001	-0.000

Table S.11: Caco-2: what replacing a model with its own probabilistic counterpart changes. Two rows per pair and noise condition, one column per representation: the change in AUC_{norm}, then the change in clean R^2, both counterpart minus base, the median of the paired differences over five scaffold folds. † marks an AUC_{norm} change with the same sign in all five. "Settings" says whether the two models share one setting.

Comparison	Settings	Condition	Change	ECFP4	PDV	MHG-GNN	Avalon	ChemBERTa	Sort & Slice
RF (300 trees) → QRF	shared	Gaussian	AUC_{norm}	+0.022	-0.004	-0.012	-0.001	+0.016†	+0.025
RF (300 trees) → QRF	shared	Gaussian	Clean R^2	-0.027	-0.033	-0.023	-0.020	-0.022	-0.042
RF (300 trees) → QRF	shared	Grouped, shifted	AUC_{norm}	-0.029†	-0.025†	-0.033†	-0.040†	-0.017	-0.028
RF (300 trees) → QRF	shared	Grouped, shifted	Clean R^2	-0.027	-0.033	-0.023	-0.020	-0.022	-0.042
RF (300 trees) → QRF	shared	Grouped, wider	AUC_{norm}	+0.048†	+0.015†	+0.009	+0.033	+0.027†	+0.040
RF (300 trees) → QRF	shared	Grouped, wider	Clean R^2	-0.027	-0.033	-0.023	-0.020	-0.022	-0.042
RF → QRF	differ	Gaussian	AUC_{norm}	-0.031	+0.029	+0.018	+0.033	+0.016	+0.010
RF → QRF	differ	Gaussian	Clean R^2	-0.021	-0.031	-0.019	-0.019	-0.024	-0.036
RF → QRF	differ	Grouped, shifted	AUC_{norm}	-0.054	-0.016	-0.056	-0.059	+0.025†	-0.043
RF → QRF	differ	Grouped, shifted	Clean R^2	-0.021	-0.031	-0.019	-0.019	-0.024	-0.036
RF → QRF	differ	Grouped, wider	AUC_{norm}	+0.038	+0.055†	+0.013	+0.046	+0.062†	+0.042

Comparison	Settings	Condition	Change	ECFP4	PDV	MHG-GNN	Avalon	ChemBERTa	Sort & Slice
RF → QRF	differ	Grouped, wider	Clean R^2	-0.021	-0.031	-0.019	-0.019	-0.024	-0.036
NN-α → BNN-α	shared	Gaussian	AUC_{norm}	+0.091	+0.016	-0.267†	+0.158	—	+0.095†
NN-α → BNN-α	shared	Gaussian	Clean R^2	-0.011	-0.102	-0.095	-0.127	—	-0.120
NN-α → BNN-α	shared	Grouped, shifted	AUC_{norm}	+0.136	+0.094	—	+0.004	—	+0.119
NN-α → BNN-α	shared	Grouped, shifted	Clean R^2	-0.030	-0.106	—	-0.110	—	-0.088
NN-α → BNN-α	shared	Grouped, wider	AUC_{norm}	-0.010	-0.061	+0.024†	-0.019	+0.111†	+0.117
NN-α → BNN-α	shared	Grouped, wider	Clean R^2	-0.027	-0.115	-0.237	-0.074	-0.269	-0.125
BNN-α → VBLL-α	differ	Gaussian	AUC_{norm}	+0.145†	-0.063	—	-0.137†	—	+0.064
BNN-α → VBLL-α	differ	Gaussian	Clean R^2	-0.146	+0.070	—	+0.024	—	-0.065
BNN-α → VBLL-α	differ	Grouped, shifted	AUC_{norm}	+0.184†	+0.185†	—	+0.274†	—	+0.191
BNN-α → VBLL-α	differ	Grouped, shifted	Clean R^2	-0.142	-0.039	—	-0.085	—	-0.032
BNN-α → VBLL-α	differ	Grouped, wider	AUC_{norm}	+0.112	+0.154	+nan	+0.378†	+0.002†	-0.140
BNN-α → VBLL-α	differ	Grouped, wider	Clean R^2	-0.152	+0.002	+nan	-0.167	+0.062	+0.026
BNN-α → BNN-α (var. head)	differ	Gaussian	AUC_{norm}	+0.014	-0.068	—	-0.276†	—	-0.139†
BNN-α → BNN-α (var. head)	differ	Gaussian	Clean R^2	-0.046	+0.015	—	+0.007	—	+0.011
BNN-α → BNN-α (var. head)	differ	Grouped, shifted	AUC_{norm}	-0.073	-0.118	—	-0.160	—	+0.031
BNN-α → BNN-α (var. head)	differ	Grouped, shifted	Clean R^2	+0.005	+0.023	—	-0.016	—	-0.008
BNN-α → BNN-α (var. head)	differ	Grouped, wider	AUC_{norm}	+0.018	-0.068	-0.686†	-0.268†	-0.487†	-0.220
BNN-α → BNN-α (var. head)	differ	Grouped, wider	Clean R^2	-0.043	+0.023	+0.125	+0.022	+0.020	-0.018
NN-β → BNN-β	shared	Gaussian	AUC_{norm}	+0.075†	-0.043	+0.119	-0.000	+0.096	-0.004
NN-β → BNN-β	shared	Gaussian	Clean R^2	-0.018	-0.035	-0.247	-0.091	-0.193	-0.035
NN-β → BNN-β	shared	Grouped, shifted	AUC_{norm}	+0.075	+0.021	+0.016†	+0.070†	+0.029	+0.138
NN-β → BNN-β	shared	Grouped, shifted	Clean R^2	-0.043	-0.021	-0.201	-0.140	-0.134	-0.075
NN-β → BNN-β	shared	Grouped, wider	AUC_{norm}	+0.048	+0.059	-0.032	+0.102	-0.125†	+0.055
NN-β → BNN-β	shared	Grouped, wider	Clean R^2	-0.016	-0.102	-0.137	-0.074	-0.070	-0.082
BNN-β → VBLL-β	differ	Gaussian	AUC_{norm}	+0.081†	+0.124†	+0.037	+0.161	+0.085†	+0.117
BNN-β → VBLL-β	differ	Gaussian	Clean R^2	-0.091	-0.017	+0.075	-0.036	+0.055	-0.051
BNN-β → VBLL-β	differ	Grouped, shifted	AUC_{norm}	+0.119†	-0.044	+0.171†	+0.209†	+0.051†	+0.132†
BNN-β → VBLL-β	differ	Grouped, shifted	Clean R^2	-0.091	+0.006	+0.022	-0.001	+0.046	-0.046
BNN-β → VBLL-β	differ	Grouped, wider	AUC_{norm}	+0.042	+0.073	+0.334†	+0.188†	+0.223†	+0.082†
BNN-β → VBLL-β	differ	Grouped, wider	Clean R^2	-0.057	-0.002	-0.063	-0.048	-0.079	-0.021
BNN-β → BNN-β (var. head)	differ	Gaussian	AUC_{norm}	-0.073	-0.082†	—	-0.135†	-0.455†	-0.052
BNN-β → BNN-β (var. head)	differ	Gaussian	Clean R^2	-0.016	-0.035	—	+0.012	+0.067	+0.007
BNN-β → BNN-β (var. head)	differ	Grouped, shifted	AUC_{norm}	-0.006	-0.088	—	-0.119†	-0.079†	-0.022
BNN-β → BNN-β (var. head)	differ	Grouped, shifted	Clean R^2	-0.060	-0.011	—	-0.085	-0.015	-0.018
BNN-β → BNN-β (var. head)	differ	Grouped, wider	AUC_{norm}	-0.095	-0.241	—	-0.190†	-0.122	-0.161†
BNN-β → BNN-β (var. head)	differ	Grouped, wider	Clean R^2	-0.028	-0.003	—	-0.061	-0.076	+0.035
GP → GP (het.)	shared	Gaussian	AUC_{norm}	+0.001	+0.002	+0.002	+0.004	+0.011	+0.005†
GP → GP (het.)	shared	Gaussian	Clean R^2	-0.000	+0.001	+0.002	-0.000	+0.002	+0.000
GP → GP (het.)	shared	Grouped, shifted	AUC_{norm}	+0.010†	+0.013†	+0.025	+0.021†	+0.055†	+0.025
GP → GP (het.)	shared	Grouped, shifted	Clean R^2	-0.000	+0.000	+0.002	-0.000	+0.002	+0.000
GP → GP (het.)	shared	Grouped, wider	AUC_{norm}	+0.002	+0.002	-0.001	+0.004	+0.003	+0.008†

Comparison	Settings	Condition	Change	ECFP4	PDV	MHG-GNN	Avalon	ChemBERTa	Sort & Slice
GP → GP (het.)	shared	Grouped, wider	Clean R^2	-0.000	+0.001	+0.002	-0.001	+0.002	+0.000

Table S.12: hERG K_i: what replacing a model with its own probabilistic counterpart changes. Two rows per pair and noise condition, one column per representation: the change in $\mathrm{AUC}_{\mathrm{norm}}$, then the change in clean R^2, both counterpart minus base, the median of the paired differences over five scaffold folds. † marks an $\mathrm{AUC}_{\mathrm{norm}}$ change with the same sign in all five. "Settings" says whether the two models share one setting.

Comparison	Settings	Condition	Change	ECFP4	PDV	MHG-GNN	Avalon	ChemBERTa	Sort & Slice
RF (300 trees) → QRF	shared	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	+0.000	-0.043†	-0.029	-0.032	-0.023	-0.017
RF (300 trees) → QRF	shared	Gaussian	Clean R^2	-0.010	+0.015	-0.007	-0.001	+0.005	-0.023
RF (300 trees) → QRF	shared	Grouped, shifted	$\mathrm{AUC}_{\mathrm{norm}}$	-0.044†	-0.073†	-0.034†	-0.058†	-0.033†	-0.047†
RF (300 trees) → QRF	shared	Grouped, shifted	Clean R^2	-0.010	+0.015	-0.007	-0.001	+0.005	-0.023
RF (300 trees) → QRF	shared	Grouped, wider	$\mathrm{AUC}_{\mathrm{norm}}$	+0.013†	-0.016†	+0.010	+0.012	-0.018	+0.005
RF (300 trees) → QRF	shared	Grouped, wider	Clean R^2	-0.010	+0.015	-0.007	-0.001	+0.005	-0.023
RF → QRF	differ	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	+0.004	-0.035†	-0.015	-0.037	-0.033	+0.015
RF → QRF	differ	Gaussian	Clean R^2	-0.009	+0.020	+0.001	+0.005	+0.019	-0.016
RF → QRF	differ	Grouped, shifted	$\mathrm{AUC}_{\mathrm{norm}}$	-0.070†	-0.054†	-0.018†	-0.056†	-0.030†	-0.026
RF → QRF	differ	Grouped, shifted	Clean R^2	-0.009	+0.020	+0.001	+0.005	+0.019	-0.016
RF → QRF	differ	Grouped, wider	$\mathrm{AUC}_{\mathrm{norm}}$	+0.031	-0.001	+0.015	+0.009	-0.006	+0.026†
RF → QRF	differ	Grouped, wider	Clean R^2	-0.009	+0.020	+0.001	+0.005	+0.019	-0.016
NN-α → BNN-α	shared	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	+0.004	+0.215†	-0.145†	+0.114	—	+0.006
NN-α → BNN-α	shared	Gaussian	Clean R^2	+0.011	-0.145	-0.266	-0.155	—	-0.057
NN-α → BNN-α	shared	Grouped, shifted	$\mathrm{AUC}_{\mathrm{norm}}$	+0.052	+0.022	-0.359†	+0.105	—	-0.030
NN-α → BNN-α	shared	Grouped, shifted	Clean R^2	+0.015	-0.158	-0.253	-0.126	—	-0.071
NN-α → BNN-α	shared	Grouped, wider	$\mathrm{AUC}_{\mathrm{norm}}$	+0.049	+0.048	+0.095†	+0.059	+0.178†	+0.126
NN-α → BNN-α	shared	Grouped, wider	Clean R^2	+0.029	-0.124	-0.334	-0.076	-0.248	-0.103
BNN-α → VBLL-α	differ	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	+0.146†	+0.142†	+0.358†	+0.088	—	+0.169
BNN-α → VBLL-α	differ	Gaussian	Clean R^2	-0.094	-0.009	+0.065	-0.007	—	+0.009
BNN-α → VBLL-α	differ	Grouped, shifted	$\mathrm{AUC}_{\mathrm{norm}}$	+0.074	+0.059	+0.590†	+0.027	—	+0.211†
BNN-α → VBLL-α	differ	Grouped, shifted	Clean R^2	-0.076	+0.025	-0.027	-0.012	—	-0.025
BNN-α → VBLL-α	differ	Grouped, wider	$\mathrm{AUC}_{\mathrm{norm}}$	+0.069†	+0.024†	+0.017	+0.135	-0.035†	-0.034
BNN-α → VBLL-α	differ	Grouped, wider	Clean R^2	-0.114	+0.029	+0.077	-0.072	+0.065	+0.005
BNN-α → BNN-α (var. head)	differ	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	-0.040	-0.625†	—	-0.384†	—	-0.324†
BNN-α → BNN-α (var. head)	differ	Gaussian	Clean R^2	-0.057	-0.052	—	-0.040	—	+0.060
BNN-α → BNN-α (var. head)	differ	Grouped, shifted	$\mathrm{AUC}_{\mathrm{norm}}$	+0.010	-0.506†	—	-0.335†	—	-0.377†
BNN-α → BNN-α (var. head)	differ	Grouped, shifted	Clean R^2	-0.029	-0.108	—	-0.029	—	+0.007
BNN-α → BNN-α (var. head)	differ	Grouped, wider	$\mathrm{AUC}_{\mathrm{norm}}$	-0.060	-0.448†	—	-0.251†	—	-0.412†
BNN-α → BNN-α (var. head)	differ	Grouped, wider	Clean R^2	-0.011	-0.055	—	-0.034	—	-0.011
NN-β → BNN-β	shared	Gaussian	$\mathrm{AUC}_{\mathrm{norm}}$	+0.008	+0.135†	+0.123†	+0.124	-0.142†	+0.003
NN-β → BNN-β	shared	Gaussian	Clean R^2	+0.076	-0.118	-0.190	-0.163	-0.215	-0.080

Comparison	Settings	Condition	Change	ECFP4	PDV	MHG-GNN	Avalon	ChemBERTa	Sort & Slice
NN-β → BNN-β	shared	Grouped, shifted	AUC_{norm}	+0.077†	-0.133†	-0.052	+0.184	—	-0.217†
NN-β → BNN-β	shared	Grouped, shifted	Clean R^2	+0.015	-0.053	-0.212	-0.049	—	-0.086
NN-β → BNN-β	shared	Grouped, wider	AUC_{norm}	+0.022	+0.055	+0.120†	+0.035	+0.030	+0.052
NN-β → BNN-β	shared	Grouped, wider	Clean R^2	+0.010	-0.157	-0.230	-0.059	-0.185	-0.093
BNN-β → VBLL-β	differ	Gaussian	AUC_{norm}	-0.009	+0.037	+0.162†	-0.103	+0.408†	+0.065
BNN-β → VBLL-β	differ	Gaussian	Clean R^2	-0.091	+0.053	+0.015	+0.078	+0.089	+0.011
BNN-β → VBLL-β	differ	Grouped, shifted	AUC_{norm}	-0.025	-0.034†	+0.250	-0.040	—	+0.268†
BNN-β → VBLL-β	differ	Grouped, shifted	Clean R^2	-0.071	+0.039	+0.039	+0.016	—	+0.035
BNN-β → VBLL-β	differ	Grouped, wider	AUC_{norm}	+0.020	-0.232	-0.049†	+0.032	+0.097†	-0.165
BNN-β → VBLL-β	differ	Grouped, wider	Clean R^2	-0.063	+0.045	+0.132	+0.034	+0.061	+0.055
BNN-β → BNN-β (var. head)	differ	Gaussian	AUC_{norm}	-0.009	-0.300†	—	-0.418†	—	-0.351
BNN-β → BNN-β (var. head)	differ	Gaussian	Clean R^2	-0.007	-0.004	—	+0.082	—	-0.054
BNN-β → BNN-β (var. head)	differ	Grouped, shifted	AUC_{norm}	-0.003	-0.156†	—	-0.444†	—	-0.143
BNN-β → BNN-β (var. head)	differ	Grouped, shifted	Clean R^2	-0.037	-0.065	—	-0.004	—	+0.003
BNN-β → BNN-β (var. head)	differ	Grouped, wider	AUC_{norm}	-0.084	-0.514†	—	-0.235†	—	-0.308†
BNN-β → BNN-β (var. head)	differ	Grouped, wider	Clean R^2	-0.014	+0.033	—	-0.071	—	+0.059
GP → GP (het.)	shared	Gaussian	AUC_{norm}	+0.001	-0.008	+0.001	+0.007	+0.010	+0.009
GP → GP (het.)	shared	Gaussian	Clean R^2	+0.000	-0.002	-0.005	-0.000	-0.010	+0.000
GP → GP (het.)	shared	Grouped, shifted	AUC_{norm}	+0.007	+0.020	+0.029	+0.030	+0.025	+0.011
GP → GP (het.)	shared	Grouped, shifted	Clean R^2	+0.000	-0.000	-0.005	-0.000	-0.011	+0.000
GP → GP (het.)	shared	Grouped, wider	AUC_{norm}	-0.002	+0.001	+0.006	+0.008	+0.017†	+0.004†
GP → GP (het.)	shared	Grouped, wider	Clean R^2	+0.000	-0.001	-0.005	-0.000	-0.012	+0.000

S.2 The flaws of the similarity assumption

One row of Table S.13 is one representation on one dataset. Representations are listed in each dataset's pool order.

Table S.13: All-pairs correlation, five-nearest-neighbor prediction and activity cliffs for every representation in each regression dataset's pool. All pairs: Spearman correlation between pair distance and absolute label difference, on a fixed sample of 2,000 molecules (seed 0) where a dataset has more. Five neighbors: Spearman correlation and RMSE (in the label's units) of the mean label of the five nearest molecules from other scaffold groups. Cliff molecules: molecules in at least one of this representation's closest pairs whose labels differ by more than the dataset's gap (Table 4.4). The number of closest pairs is the number ECFP4 has at Tanimoto ≥ 0.55, a few more where distances tie at the cut-off. Shared: percentage of ECFP4's similar pairs that are also among this representation's closest pairs. HLM clearance's pool has no ECFP4.

Dataset	Representation	All pairs Spearman	Five neighbors Spearman	Five neighbors RMSE	Cliff molecules	Shared (%)
hERG K_i	MHG-GNN	0.07	0.71	0.65	208	67
	Mordred	0.07	0.72	0.62	165	62
	WL graph kernel	0.04	0.71	0.64	267	74
	Avalon	0.02	0.69	0.65	238	63
	GROVER	0.05	0.70	0.63	192	63
	PDV	0.08	0.73	0.62	208	61
	ECFP4	0.06	0.72	0.63	211	100
	Atom pairs	0.04	0.74	0.60	138	68
	ChemBERT	0.07	0.68	0.66	166	57
	SMI-TED	0.09	0.69	0.65	187	58
LogD	MHG-GNN	0.09	0.81	0.66	286	42
	Mordred	0.06	0.80	0.68	308	35
	GROVER	0.09	0.81	0.67	399	51
	ECFP4	0.12	0.76	0.74	731	100
	SMI-TED	0.10	0.76	0.74	758	42
	MoLFormer	0.13	0.77	0.73	768	46
	Avalon	0.04	0.73	0.77	801	49
	Topological torsions	0.10	0.74	0.76	808	71
	ChemBERT	0.09	0.75	0.75	808	42
	GraphMVP	0.11	0.76	0.74	415	38
Caco-2 efflux	WL graph kernel	0.13	0.58	0.35	398	54
	MHG-GNN	0.16	0.60	0.34	347	53
	ECFP4	0.08	0.59	0.34	324	100
	PDV	0.18	0.62	0.34	237	43
	GROVER	0.06	0.61	0.34	302	55
	Avalon	−0.00	0.56	0.35	335	46
	Topological torsions	0.15	0.61	0.34	348	74
	SMI-TED	0.14	0.57	0.35	400	44
	Atom pairs	0.08	0.59	0.34	315	57
	MoLFormer	0.09	0.59	0.34	311	51
RLM clearance	PDV	0.06	0.56	0.62	433	34

Dataset	Representation	All pairs Spearman	Five neighbors Spearman	Five neighbors RMSE	Cliff molecules	Shared (%)
	MHG-GNN	0.02	0.47	0.67	487	46
	SMI-TED	0.04	0.46	0.67	483	48
	Mordred	0.07	0.49	0.66	445	42
	MoLFormer	0.08	0.44	0.68	412	60
	GROVER	0.05	0.43	0.69	436	46
	WL graph kernel	0.00	0.40	0.71	554	36
	Atom pairs	0.07	0.47	0.67	437	60
	Avalon	0.04	0.29	0.74	389	55
	ECFP4	0.02	0.36	0.71	458	100
MDR1–MDCK efflux	Mordred	0.16	0.58	0.57	363	45
	PDV	0.23	0.64	0.54	302	39
	MHG-GNN	0.20	0.52	0.62	344	48
	Avalon	−0.04	0.50	0.61	378	61
	GROVER	−0.03	0.47	0.63	356	49
	SMI-TED	0.10	0.59	0.56	389	51
	Atom pairs	0.00	0.58	0.57	374	64
	WL graph kernel	0.17	0.47	0.64	413	40
	MACCS keys	−0.04	0.54	0.59	379	46
	ECFP4	0.02	0.51	0.61	397	100
HLM clearance	Mordred	0.07	0.49	0.55	429	43
	PDV	0.10	0.55	0.52	390	34
	MHG-GNN	0.08	0.46	0.56	435	46
	SMI-TED	0.03	0.45	0.56	475	48
	MoLFormer	0.00	0.43	0.56	422	60
	GROVER	−0.03	0.41	0.59	413	46
	Atom pairs	−0.04	0.47	0.55	426	60
	ChemBERT	−0.02	0.39	0.58	452	45
	WL graph kernel	0.12	0.38	0.59	490	37
	Avalon	0.01	0.32	0.61	416	56
QM9	Mordred	0.18	0.84	0.68	1,443	3
	Avalon	0.19	0.83	0.70	1,148	9
	SMI-TED	0.13	0.85	0.65	1,576	13
	ECFP4	0.19	0.88	0.64	1,495	100
	SELFormer	0.20	0.82	0.73	300	2
	ChemBERTa-2	0.19	0.89	0.61	1,108	1
	Mol2vec	0.14	0.86	0.66	1,274	24
	ChemBERT	0.19	0.83	0.72	1,295	7
	GraphMVP	0.20	0.83	0.71	1,231	9
	MoLFormer	0.21	0.84	0.70	1,423	3

S.3 Building adaptive hybrid representations

Additional file 10: Settings of every model, as downstream model and as internal evaluator

One row is one model. The middle two columns give its settings when it scores a hybrid (downstream model) and when it scores one representation inside the search (internal evaluator). The models fitted only to compute feature importance are described in Chapter 5.

Table S.14: Settings of every model. p is the number of features. "Molecules" is the number of training molecules the internal evaluator is fitted on; above it, a random subset of that size is drawn with a fixed seed. Settings not listed are the library's defaults. Every setting is the same on every dataset. The Gaussian process runs only on the datasets named in Chapter 5, "Models and metrics".

Model	As downstream model	As internal evaluator	Molecules	Library; source
Random forest	100 trees, no depth limit; $\sqrt{p}$ features tried at each split	50 trees, depth at most 10; all p features tried at each split on regression, $\sqrt{p}$ on classification	all	scikit-learn [1]; Breiman [2]
LightGBM	100 trees	50 trees, depth at most 4	all	lightgbm; Ke et al. [3]
XGBoost	100 trees; half of the features sampled for each tree	50 trees, depth at most 4	all	xgboost; Chen and Guestrin [4]
SVM	RBF kernel, $C = 1$. Not fitted above 10,000 molecules, or above 5,000 under uncapped augmentation	RBF kernel, $C = 1$	at most 500	scikit-learn; Vapnik [5]
Gaussian process	Regression: RBF plus white-noise kernel, labels normalized, 3 optimizer restarts. Classification: RBF kernel. Not fitted above 1,000 molecules	Regression: scaled RBF kernel, fixed noise variance 0.01, no restarts. Classification: scaled RBF kernel, no restarts	at most 500	scikit-learn; Rasmussen and Williams [6]

Model	As downstream model	As internal evaluator	Molecules	Library; source
DNN	Hidden layers of 256, 128 and 64 units, ReLU, dropout 0.2 [7]; Adam [8] at learning rate 0.001, halved after 5 epochs without improvement; batch size 256; up to 200 epochs; stops after 20 epochs without improvement on 20% of the training molecules that share no group with the rest, using the outer split's grouping; labels z-normalized	The same, except that the early-stopping 20% is drawn at random	all	PyTorch [9]

Additional file 11: Downstream models left out, by dataset and augmentation setting

One row is one rule that leaves a downstream model out. Only the 16 datasets of Table 5.8 are listed; a dataset and setting not named in any row runs every downstream model in Table 5.6.

Table S.15: Downstream models left out, by dataset and augmentation setting. Every rule was set so that a search finishes within the time limit of one job; none was set from scores. hERG K_i drops no model under any setting.

Model left out	Augmentation setting	Datasets
XGBoost	all three	H3K4me3, TATA promoter
SVM	all three	datasets above 10,000 molecules: H3K4me3
SVM	uncapped	datasets above 5,000 molecules: QM9, LogD[a], TATA promoter[a]
SVM	uncapped	Koenig expression
SVM	capped	Caco-2 efflux, LogD, rat liver microsome clearance, MDR1–MDCK efflux, Buchwald–Hartwig plates
Random forest and SVM	capped	QM9
Random forest	uncapped	QM9, LogD
Gaussian process	all three	every dataset except humanness (Chapter 5, "Models and metrics")

[a]Counted on the whole dataset: LogD has 5,028 molecules and TATA promoter 5,274. A

search run inside one outer fold is given about 80% of them and keeps the SVM.

Additional file 12: Size of the search

One row is one downstream model. The counts are per dataset, objective and augmentation setting, with adaptive pruning off.

Table S.16: Number of configurations searched per dataset, objective and augmentation setting. A configuration is one choice of downstream model, internal evaluator, allocation method, importance method, representation count and budget. At the full tier each (downstream model, internal evaluator) pair runs 5 allocation methods × 3 importance methods × 4 representation counts × 8 budgets = 480 configurations; at the fast tier it runs $3 \times 1 \times 2 \times 8 = 48$ (Table 5.6). "Matched" is counted once where it repeats an evaluator already in the list. Each configuration is scored on 5 folds, so the full tier fits a downstream model 40,800 times (50,400 on humanness) and the fast tier 4,080 times. Where a dataset leaves a downstream model out (Additional file 11), its pairs drop out: a fast search on H3K4me3 has 9 pairs, so 432 configurations.

Downstream model	Internal evaluators	Full tier	Fast tier
Random forest	3 (matched is random forest)	1,440	144
LightGBM	3 (matched is LightGBM)	1,440	144
DNN	3 (matched is DNN)	1,440	144
XGBoost	4	1,920	192
SVM	4	1,920	192
Gaussian process (humanness only)	4	1,920	not run
Total	17 pairs (21 on humanness)	8,160 (10,080 on humanness)	816

Additional file 13: Design experiments behind the search settings

One row is one setting of the search, or one rule that was tested and then fixed instead of searched. Every number comes from preliminary design runs made before the main experiments; they are not results of the thesis, and several used other budgets than the final search.

Table S.17: Design experiments behind the search settings. Preliminary design runs, not the thesis's full runs.

Setting	Values compared	Datasets, folds, budgets	What was measured, and the result	Values kept (full / fast)
Downstream model	Nine models, among them random forest, LightGBM, XGBoost, CatBoost, DNN, k-nearest neighbors and a Gaussian process, each on each of 10 base representations at full width	Buchwald–Hartwig yield, Caco-2 efflux, Koenig expression, LogD, breast cancer PAM50 subtype; 5 folds	Best held-out score over the 10 representations, and seconds per fit. The DNN scored best on 4 of the 5 datasets and third on Caco-2 (0.2936 against 0.2851 RMSE, $\log_{10}$ efflux ratio)	Full: random forest, LightGBM, XGBoost, SVM, DNN, and the Gaussian process on humanness / fast: the same five
Budget	None compared	–	–	10, 25, 50, 100, 250, 500, 1,000 and 2,000 features at both tiers
Representation count	2, 3, 5 and all 10	9 datasets, 3 tree-ensemble downstream models, budgets 500 and 1,000 features; 5 folds	The best count for each (downstream model, budget) combination	Full: 2, 3, 5, all / fast: 3, all
Allocation method	8 candidates: the five searched and 3 further pooled candidates, each at every representation count	The same 9 datasets, models and budgets: 54 (dataset, model, budget) combinations	Each candidate was scored at its best representation count, and the number of combinations in which it was in the top 3 of the 8 was counted. Fixed shares: linear 30, squared 27, rank decay 26, softmax 25; global pool with squared weights 14	Full: the five described in Chapter 5 / fast: linear, squared, rank decay

Setting	Values compared	Datasets, folds, budgets	What was measured, and the result	Values kept (full / fast)
Importance method, agreement	8 methods	LogD, hERG K_i, Buchwald–Hartwig yield, nanobody thermostability, antibody FcRn retention, breast cancer PAM50 subtype, lung adenocarcinoma vs squamous; each method fitted once on the whole dataset	Spearman correlation between two methods' scores over every feature. Kept pairs: LightGBM gain and TreeSHAP 0.20–0.64, LightGBM gain and DeepLIFT 0.21–0.67, TreeSHAP and DeepLIFT 0.56–0.99 across the 7 datasets	One method kept from each group whose rankings agree above 0.95
Importance method, score	28 methods, among them three controls: feature variance, Laplacian score and random ranking	Caco-2 efflux; 3 folds; budget 100 features; random forest and LightGBM downstream	RMSE of the best single representation minus RMSE of the hybrid ($\log_{10}$ efflux ratio; positive means the hybrid has the lower error): DeepLIFT +0.0144, LightGBM gain +0.0100, TreeSHAP +0.0067, random ranking −0.0118	Full: LightGBM gain, TreeSHAP, DeepLIFT / fast: LightGBM gain
Internal evaluator settings	20 hyperparameter settings from 5 model families	Buchwald–Hartwig yield, Caco-2 efflux, Koenig expression; 3 folds; budget 100 features	Spread of mean scores across all 20 settings: 0.0253 R^2, 0.0179 R^2 and 0.0085 Spearman. Median standard deviation across the 3 folds for one setting, allocation method and downstream model: 0.0462 R^2, 0.0693 R^2 and 0.0186 Spearman	Matched, random forest, LightGBM and DNN, at fixed settings, both tiers

Setting	Values compared	Datasets, folds, budgets	What was measured, and the result	Values kept (full / fast)
Rescaling of each representation's score (a rule)	5 rescalings	9 datasets; 5 folds; budgets 50 and 300 features	Whether any rescaling beat the one in use by more than that rule's own spread across folds: in 0 of 90 (dataset, allocation method, budget) combinations	One fixed rule per metric (Chapter 5)

The representation-count and allocation rows come from one design run on 10 datasets, 3 downstream models and 2 budgets (60 combinations); QM9 is left out, which gives the 54.

Values of the search settings. Before the full runs, we ran a series of preliminary design experiments to determine the values each setting of the search may take. They used 1 to 9 datasets and 1 to 5 folds, and each dataset was scored in its own metric, with scores never averaged across datasets. When making a selection, we considered whether a value gave the best score for some dataset, downstream model and budget, as well as the compute time in hours. A value that won somewhere but cost many hours stayed in the full search and was left out of the fast tier. Each setting's values were then either kept, because each won for some dataset, model and budget, or fixed at one value. A setting was fixed when the spread between its values was smaller than the spread of one value across folds. As an example of a kept value, nine downstream models were fitted on each of 10 base representations over 5 folds. The datasets were Buchwald–Hartwig yield, Caco-2 efflux, Koenig expression, LogD and breast cancer PAM50 subtype. The DNN scored best on 4 of the 5 datasets and third on Caco-2 (0.2936 against 0.2851 RMSE, $\log_{10}$ efflux ratio), so it was added. As an example of a fixed rule, we tested five ways of rescaling each representation's score before the budget is split. On hERG K_i, with linear allocation at 300 features, the five rescalings gave RMSE values from 0.5870 to 0.5924 ($\mathrm{p}K_i$). A single rescaling, however, ranged from 0.5475 to 0.6294 RMSE across the 5 folds, so rescaling is one fixed rule per metric. Every design experiment is listed in Table S.17.

Importance methods. Two comparisons were run, the first of how the methods rank features and the second of the hybrids those rankings build. The ranking comparison scored 8 importance methods on 7 datasets, namely LogD, hERG

K_i, Buchwald–Hartwig yield, nanobody thermostability, antibody FcRn retention, breast invasive carcinoma (BRCA) PAM50 subtype and lung adenocarcinoma (LUAD) vs lung squamous cell carcinoma (LUSC). For each pair of methods, it measured the Spearman correlation between their scores over every feature of every representation. The three retained methods rank features differently. Across the 7 datasets, the correlation ranged from 0.20 to 0.64 between LightGBM gain and TreeSHAP, from 0.21 to 0.67 between LightGBM gain and DeepLIFT, and from 0.56 to 0.99 between TreeSHAP and DeepLIFT. These numbers are preliminary, because each method was fitted once on the whole dataset rather than inside folds.

The hybrid comparison used Caco-2, with 2,158 molecules, 10 representations, 3 scaffold folds and a budget of 100 features. Hybrids were built from each method's ranking and scored with random forest and LightGBM downstream models. Three control rankings were added, namely feature variance, Laplacian score and a random order. The measure was the RMSE of the best single representation minus the RMSE of the hybrid, in $\log_{10}$ efflux ratio units, so that a positive value means the hybrid has the lower error. The preliminary results were $+0.0144$ for DeepLIFT, $+0.0100$ for LightGBM gain, $+0.0067$ for TreeSHAP and -0.0118 for the random ranking. The gap between DeepLIFT and LightGBM gain, 0.0044 RMSE, is smaller than the fold-to-fold spread. The methods also differ at the top of their rankings. Of each representation's 100 top-ranked features on Caco-2, the retained pairs of methods share a median of 28 to 54 features.

Three rules determined which methods were retained. First, a method must score every feature of every representation for regression, binary and multiclass labels, which all three retained methods do. Second, of two methods whose rankings agree above a Spearman correlation of 0.95, we keep the faster one. Under this rule, DeepLIFT replaced integrated gradients, which agreed with it above 0.95 on 4 of 6 datasets, and gradient SHAP, which agreed above 0.95 on 3 of 6. Third, we keep methods that rank features differently, because a different ranking builds a different hybrid. TreeSHAP came last of the three on Caco-2 and was kept under this third rule. Only two filters act on features before allocation, the cut at a score of 0 and the pre-screen on representations wider than 2,000 features. On a 2,048-bit fingerprint, the pre-screen drops the 48 lowest-variance bits. On hERG K_i all 48 are among the 105 constant ECFP4 bits, while on QM9, which has no constant ECFP4 bits, they are near-constant ones. Near-constant features otherwise remain candidates and are ranked by importance.

No single importance method suits every dataset and downstream model. A preliminary run used 5 datasets (nanobody thermostability, hERG K_i, humanness, LogD and BRCA PAM50 subtype), 5 folds, 10 representations, 5 downstream models and budgets of 25, 50, 100, 150 and 300 features. For each of the 25 pairs of downstream model and budget per dataset, we counted which importance method gave the best configuration, and Table S.18 gives the counts. LightGBM gain won the most pairs on three datasets and tied with TreeSHAP on humanness. DeepLIFT won the most on one dataset, LogD, with 17 of its 25 pairs, and outside LogD it gave the best configuration for 2 of 100 pairs. With the DNN downstream model, LightGBM gain won 17 of 20 budgets outside LogD, whereas on LogD the DNN took DeepLIFT at 3 of 5 budgets. On LogD, XGBoost and LightGBM each took DeepLIFT at 5 of 5 budgets, and random forest took LightGBM gain at 5 of 5. **[RESULT NEEDED: the same win counts, per dataset and importance method, from the full runs]**

		Wins (of 25)			Best score		
Dataset	Metric	LightGBM gain	TreeSHAP	DeepLIFT	LightGBM gain	TreeSHAP	DeepLIFT
Nanobody thermostability	Spearman	15	10	0	0.6696	0.6471	0.6440
hERG K_i	RMSE (pK_i)	14	11	0	0.5589	0.5605	0.5728
Humanness	AUROC	12	12	1	0.9723	0.9713	0.9689
LogD	RMSE (log units)	8	0	17	0.4250	0.4345	0.4170
BRCA PAM50 subtype	weighted F1	21	3	1	0.9097	0.9018	0.9038

Table S.18: Importance method of the best configuration in a preliminary run on five datasets, with 5 folds, 10 representations, 5 downstream models and budgets of 25, 50, 100, 150 and 300 features. Wins count the pairs of downstream model and budget, out of 25 per dataset, at which a method gave the best configuration, and Best score is the best score reached with that method.

Allocation methods. We first tried greedy forward selection, which picks features for how well they work together, something fixed shares cannot do [10, 11]. In each round, it added the next block of features from whichever representation most improved the internal validation score. It stopped when the budget was spent or when the score had not improved for a set number of rounds. With blocks of s features and K representations, it needs up to $(B/s) \times K$ fits per configuration and fold. With blocks of 10 features, this amounts to 500 fits at 500 features and 10 representations, against 10 for the score-based methods. In preliminary runs, it reached a Spearman correlation of 0.3600 on Jain antibody FcRn retention, against 0.4526 for linear allocation, and placed 16th of 17 methods. On breast cancer PAM50

subtype, it reached an accuracy of 0.8551 against 0.8938 for linear allocation, again placing 16th of 17, and it was removed from the search.

The five methods were chosen in a preliminary comparison run before the main experiments. It compared 8 candidate allocation methods, each at 4 representation counts, on 9 regression datasets. It used 3 downstream models (LightGBM, random forest and CatBoost) and budgets of 500 and 1,000 features, giving 54 combinations of dataset, downstream model and budget. A fixed-share method was the single best method in 38 of the 54 combinations and the global pool in 16. All five were kept because no single method was best in most combinations. This comparison predates the current score rescaling, and it sized the weights with an inner 3-fold group split instead of the 80/20 internal split used now.

Matched internal evaluator. Matching the evaluator to the downstream model is the obvious default. However, the evaluator only has to rank representations, and another model may rank them better for the downstream model. Because the matched evaluator is one of the four searched values, the search already covers this default, and a mismatched evaluator winning is informative rather than a negative result. We first counted wins in the search itself, in a preliminary run on 5 datasets (nanobody thermostability, hERG K_i, humanness, LogD and breast cancer PAM50 subtype). These runs used 10 representations, budgets of 25 to 300 features and 5 folds, and their scores are the search's own fold scores rather than held-out scores. The matched evaluator gave the best score for its downstream model in 9 of 25 pairs of dataset and downstream model. Counted per budget, it was best in 43 of 125 combinations of dataset, downstream model and budget. With 3 or 4 evaluators per downstream model and no effect of matching, about 38 of 125 would be expected. On nanobody thermostability with a DNN downstream model, the matched evaluator reached a Spearman correlation of 0.6696 against 0.6506 for the LightGBM evaluator, a gap of 0.0190. Its 5 fold scores ranged from 0.5095 to 0.7920. On LogD with an SVM downstream model, the matched evaluator scored 0.4322 RMSE (log units) against 0.4217 for the LightGBM evaluator, where lower is better. This is a gap of 0.0105, the matched evaluator lost at all 5 budgets, and its 5 fold scores ranged from 0.4118 to 0.4634. We then scored the evaluators on held-out folds in a second preliminary run, with 3 outer folds, a budget of 250 features, 5 representations and LightGBM gain only. The downstream models were random forest and SVM. On nanobody thermostability with a random forest downstream model, the

matched evaluator reached a Spearman correlation of 0.6098, the LightGBM evaluator 0.6220 and the SVM evaluator 0.6172. The matched evaluator's fold scores ranged from 0.5659 to 0.6734. With an SVM downstream model, the matched evaluator reached 0.6189, the Gaussian process evaluator 0.6265 and the LightGBM evaluator 0.6317, with fold scores for the matched evaluator from 0.5530 to 0.6703. On hERG K_i with a random forest downstream model, the matched evaluator scored 0.6046 RMSE ($\mathrm{p}K_i$), the SVM evaluator 0.6059 and the LightGBM evaluator 0.6075. Its fold scores ranged from 0.5957 to 0.6178, and it beat the SVM evaluator on 2 of 3 folds. With an SVM downstream model, the matched evaluator scored 0.5723 RMSE ($\mathrm{p}K_i$), the LightGBM evaluator 0.5724 and the Gaussian process evaluator 0.5765, with fold scores for the matched evaluator from 0.5605 to 0.5862. Every gap above, both held out and within the search, is smaller than the range of the matched evaluator's own fold scores. **[RESULT NEEDED: per dataset and downstream model, how often the winning configuration used the matched evaluator, and its gap to the best other evaluator in the dataset's metric]**

Fast tier. The fast lists were chosen by one rule in a preliminary comparison. A value stays in the fast lists if it is inexpensive and improves the best score, whereas an expensive value that wins often stays in the full search only. Importance methods and downstream models were measured on five datasets, with 5 folds each, against the unpruned full search. The five were breast cancer PAM50 subtype, nanobody thermostability, humanness, hERG K_i and LogD. DeepLIFT took about half of the fast run's hours on all five datasets (49.9% to 51.6%), and dropping it changed only LogD's best score, from 0.4905 to 0.4930 RMSE (log units). Adding the DNN as a downstream model moved LogD from 0.4930 to 0.4273 RMSE, against a full-search best of 0.4170. Adding the SVM moved hERG K_i from 0.5776 to 0.5599 RMSE ($\mathrm{p}K_i$), while adding XGBoost changed no dataset's best score. All five downstream models are now searched at both tiers. The global pool won 60 of the 125 combinations of dataset, downstream model and budget on those five datasets. It is expensive, however, so it is in the full search only. The three fast allocation methods are linear, squared and rank-decay allocation. **[RESULT NEEDED: the share of run time the fast tier and adaptive pruning saved on the full runs]**

Additional file 14: Fields a blueprint saves

One row is one field. The first block is the recipe that rebuilds a hybrid; the second is kept for inspection and is not read by a rebuild.

Table S.19: Fields a blueprint saves. The stored score of each representation is its best score on its own across the downstream models, over every molecule the search ran on. It is not the internal-evaluator score that splits the budget, which a rebuild computes again on the molecules it is given. Two further fields are read only by allocation or augmentation routes outside the thesis's search and are fixed: a minimum improvement a whole block of extra features must show (none) and a number of features taken from each source (20).

Field	What it holds
The recipe	
Representations	The representations offered to the search
Representation count	How many of them the configuration used; empty when it used all
Allocation method	One of the five described in Chapter 5
Importance method	LightGBM gain, TreeSHAP or DeepLIFT
Budget	Number of features in the hybrid
Internal evaluator	Matched, random forest, LightGBM or DNN
Objective	Predictive, or predicting each molecule's activity from its five nearest neighbors
Metric	The dataset's metric, and whether lower is better
Task type	Regression or classification
Folds and seed	The number of folds, and the random seed
The recipe: augmentation	
Augmentation setting	None, uncapped or capped
Candidate ranking	How candidate extra features are ranked
Candidate list length	50 features
Cap per source	Optional; the most candidates taken from one source
Margin	How much better than adding none a block of extra features must score
Fraction of the budget	Optional; used instead of the fixed block sizes
Kept for inspection, not read by a rebuild	
Representation scores	Each representation's best score on its own across the downstream models, sign flipped for metrics where lower is better
Features per representation	The number of features each representation got in each fold
Extra features kept	The extra features augmentation kept in each fold, by source

Additional file 15: Reduction of wide representations when the comparator is chosen

A representation wider than 5,000 features is first reduced, inside each fold, to the 500 features on which a ranking model splits most often. It keeps 200 features instead when the search is given fewer than 1,000 molecules. The ranking model

is a LightGBM model of 50 trees of depth at most 5, fitted on the fold's training molecules. It ranks features by how many times its trees split on them rather than by gain. This reduction applies only when the search scores a single representation on its own.

Additional file 16: Worked example of fold counts

When a training fold holds fewer than five groups, both splits use as many folds as there are groups. For example, Koenig expression has five groups, so each outer training fold holds four. The search inside that fold uses 4 folds of three groups each, and the internal split holds out one of the three.

S.4 Hybrid representations for similarity and label noise

Additional file 17: Settings of the twelve LNL methods and the plain network they share

One row is one method; the first row is the plain network every method starts from. The kind of each method follows Chapter 6.

Table S.20: Settings of the twelve LNL methods. On classification datasets the squared-error loss is replaced by cross-entropy.

Method	Kind	Settings	Labels
Plain network	–	Hidden layers of 256, 128 and 64 units, ReLU, dropout 0.2; Adam at learning rate 0.001, halved after 5 epochs without improvement; batch size 256; gradient norm clipped at 1; up to 200 epochs; stops after 20 epochs without improvement on 20% of the training molecules held out by scaffold group (drawn at random after a method has removed molecules); labels z-normalized	both
Small-loss selection	noise mitigation	10 warm-up epochs on all molecules; keeps the 70% with the smallest loss and fits a new plain network on them	both
Confident learning	noise mitigation	3 plain networks; removes molecules whose error is above the 85th percentile and whose spread across the 3 networks is below the 25th percentile, then refits	both

Method	Kind	Settings	Labels
Removal by distance to neighbors	noise mitigation	10 nearest neighbors in the standardized features; score is the distance of a label from its neighbors' mean label, divided by their standard deviation (on classification, the share of neighbors with another label); removes molecules above the 85th percentile, then refits	both
Staged removal	noise mitigation	3 stages; each removes the 10% of the remaining molecules with the largest training loss	both
Per-epoch down-weighting	noise-robust training	After each epoch, the weight of every molecule whose loss is above the median is multiplied by 0.95	both
Curriculum by confidence	noise-robust training	20 warm-up epochs, then trains on the 25%, 50%, 75% and 100% of molecules with the smallest loss, in stages	both
DivideMix	noise-robust training	Two networks, 10 warm-up epochs; a two-component Gaussian mixture on each network's squared errors splits the molecules	both
Meta-Weight-Net	noise-robust training	The weighting network is updated every 5 epochs at learning rate 0.001, on 20% of the training molecules held out for it	both
MentorNet	noise-robust training	The mentor network reads each molecule's current loss, its last 5 losses and its mean distance to its 10 nearest neighbors	both
Contrast to Divide	noise-robust training	5 perturbed copies of each molecule, Gaussian noise of standard deviation 0.01 on the features	both
Heteroscedastic GP	noise-robust training	RBF kernel, 2 optimizer restarts, fitted on at most 1,000 training molecules; a small network predicts each molecule's noise variance	regression
Deep evidential regression	noise-robust training	Weight of the evidence penalty 0.01	regression

S.5 Datasets and curation

This appendix gives, for every dataset, its source, its label and each curation step in the order the steps run. Chapter 5 and Table 5.8 give the labels, metrics, fold groupings and splitting rules of its 16 datasets. Table S.21 gives the sample counts before and after curation, obtained by running the curation code on the downloaded files.

Dataset	Released	Used	Steps that remove samples
Small molecules			
hERG K_i	1,815	1,412	no pChEMBL value (3); not a binding assay (75); repeats merged into one median per compound (322); InChIKey duplicates (3)
LogD	5,326	5,028	no value (287); repeats merged after standardization (11)
Caco-2 efflux	5,326	2,158	no value (3,165); repeats merged after standardization (3)
Rat liver microsome clearance	3,521	3,053	no value (467); InChIKey duplicates (1)
MDR1–MDCK efflux	3,521	2,641	no value (879); InChIKey duplicates (1)
QM9	130,831	9,978	not parsable by RDKit (1,403); random draw of 10,000; InChIKey duplicates (22)
hERG blockers, training set	7,772	7,606	also in the hERG K_i set (166)
hERG blockers, test set	4,383	4,213	also in the hERG K_i set (170)
hERG IC_{50}	12,010	12,008	missing or non-positive value (2)
Chemical reactions			
Buchwald–Hartwig yield	3,955	3,955	none
Buchwald–Hartwig plates	4,599	4,312	no aryl halide (287)
Antibodies			
Koenig expression	4,275	4,275	none
Nanobody thermostability	672	672	none
Humanness	564	564	none
Trastuzumab–HER2 binding	524,346	5,000	stratified random draw of 5,000
Nucleotide sequences			
H3K4me3	18,244	18,244	none (training and test files merged)
TATA promoter	5,274	5,274	none (training and test files merged)
Cancer genomics (TCGA)			
Lung adenocarcinoma vs squamous	1,129	1,017	not a primary tumor (112)
Breast cancer PAM50 subtype	1,097	821	no PAM50 call, or called normal-like (276); count is of primary tumors

Table S.21: Sample counts of each dataset before and after curation. Released is the number of rows in the downloaded files (for breast cancer, of primary tumors with expression data), and Used the number left after curation. The last column lists the steps that remove samples, with the number removed where it applies.

S.5.1 Small Molecules

Three steps are shared by several small-molecule datasets. The first is structure standardization with RDKit [12]: a general cleanup, keeping the largest fragment, neutralizing charges and selecting the canonical tautomer, in that order. Molecules that then contain a metal atom are rejected. Standardization is applied to hERG K_i, LogD and Caco-2; the Biogen sets, QM9 and the hERG blocker sets are only checked to parse in RDKit.

The second removes duplicate molecules by InChIKey [13], keeping the first occurrence, so that one molecule cannot sit in both a training and a test fold. It is applied to hERG K_i, LogD, Caco-2, the two Biogen sets and QM9, and to the hERG blocker sets only across sites (below). The third is fold grouping by Bemis–Murcko scaffold [14] (Chapter 5).

hERG K_i. hERG records (ChEMBL target CHEMBL240) were retrieved from ChEMBL [15], keeping binding assays (assay type B) with standard type K_i, a pChEMBL value, an exact relation ("=") and no curator data-validity comment. The label is the pChEMBL value, $\mathrm{p}K_i = -\log_{10} K_i$ with K_i in molar units. Repeat measurements of one compound are merged into their median, and compounds whose repeats have a standard deviation above 1.0 $\mathrm{p}K_i$ unit are removed, following Landrum and Riniker [16]. This gives 1,415 compounds, 226 of them measured more than once and none removed by the standard-deviation rule. After standardization, InChIKey deduplication removes 3, leaving 1,412 compounds with $\mathrm{p}K_i$ from 4.07 to 9.41. Because repeats are merged on ChEMBL's own SMILES before standardization, two ChEMBL entries that standardize to the same molecule are not merged by median; the InChIKey step keeps the first of them. The file read by the experiments was pulled on 1 February 2026; the ChEMBL release was not recorded at the time. Re-running the same query in September 2026 against ChEMBL release 37 gives a byte-identical file of 1,415 compounds.

LogD and Caco-2 efflux. Both come from the training split of the OpenADMET–ExpansionRx challenge data on Hugging Face [17]. For each endpoint, rows without a value are dropped, structures are standardized, and rows that standardize to the same SMILES are merged into their median label. The unitless efflux ratio is transformed to $\log_{10}$. Labels are clipped at the 1st and 99th percentiles of each dataset, -1.0 and 4.5 for LogD and -0.319 and 1.679 for the

$\log_{10}$ efflux ratio. This sets 116 LogD and 45 Caco-2 labels to a limit before any noise is added in Chapter 6. InChIKey deduplication removes nothing.

Rat liver microsome clearance and MDR1–MDCK efflux. These are two endpoints of the Biogen ADME set of 3,521 compounds [18], from Polaris as `biogen/adme-fang-v1` [19]: intrinsic clearance $\mathrm{CL}_{\mathrm{int}}$ in mL/min/kg and an efflux ratio, both released and used as $\log_{10}$ values. Rows without a value, or whose SMILES RDKit cannot parse, are dropped, and InChIKey deduplication removes one molecule from each. There is no standardization, median merge or clipping.

QM9. QM9 [20, 21] was read through PyTorch Geometric [22], which distributes 130,831 molecules. Of these, 129,428 parse in RDKit. From these, 10,000 molecules were drawn once at random with seed 42, so the subsample is a random draw of molecules, not a random split. InChIKey deduplication removes 22, leaving 9,978. The label is the HOMO–LUMO gap in eV, from 1.88 to 11.37 eV in this sample.

hERG blockers held at separate sites. The separate-site experiments of Chapter 5 use two further hERG sets from Lhasa Limited [23, 24]: a training set of 7,772 molecules (3,630 blockers, 46.7%) and a test set of 4,383 molecules (1,084 blockers). Each label is Lhasa's own binary blocker call, made under different rules in the two sets. Both are proprietary (Additional file S.6). The hERG K_i set acts as a third site, with a compound counted as a blocker when $\mathrm{p}K_i \geq 6.0$, giving 467 blockers and 945 non-blockers. Molecules are matched by InChIKey across the sites, in the order hERG K_i, training set, test set, keeping the first copy. This removes 166 training and 170 test molecules, all of them also in the hERG K_i set; the two Lhasa sets share none.

hERG IC_{50}. This set is used only in Chapter 6, where its repeat measurements and curator flags serve as real evidence of label error. It keeps every measurement as its own row. The query takes ChEMBL records for CHEMBL240 with standard type IC_{50} in nM and an exact relation [15], dropping records with a missing or non-positive value. The label is $\mathrm{pIC}_{50} = 9 - \log_{10}(\mathrm{IC}_{50}/\mathrm{nM})$. Assay type is not filtered. Of the 12,010 records, 10,219 come from binding assays, 1,543 from toxicity assays, 217 from functional assays and 31 from ADME assays, as ChEMBL classifies them. The 301 records with a curator data-validity comment, on 215 molecules, are kept

and marked. With no standardization, median merge or InChIKey step, the result is 12,008 measurements of 9,802 molecules. Folds are grouped by scaffold, so no molecule is split between training and test.

S.5.2 The Chapter 3 versions of LogD, Caco-2, hERG K_i and QM9

Chapter 3 used the same sources as Chapter 5 for LogD, Caco-2 and hERG K_i but a lighter curation, which explains the different sample counts (Table S.22). Its QM9 replicates are drawn from the same 129,428 parsable molecules (Section 3.2). It also merges hERG K_i repeats a second time, on its own standardized SMILES; this merges nothing, so its set keeps 1,415 compounds.

	Chapter 3	Chapter 5
Structure standardization	largest fragment and canonical SMILES only	cleanup, largest fragment, neutralized charges, canonical tautomer, metals rejected
Label clipping (LogD, Caco-2)	none	1st and 99th percentile
InChIKey deduplication	none	first occurrence kept
Scaffold key	stereochemistry removed; each acyclic molecule its own group	stereochemistry kept; all acyclic molecules one group
Molecules: LogD, Caco-2, hERG K_i	5,039; 2,161; 1,415	5,028; 2,158; 1,412

Table S.22: Curation of LogD, Caco-2 and hERG K_i in Chapters 3 and 5. The last row gives the numbers of molecules.

S.5.3 Chemical Reactions

Both reaction datasets come from the high-throughput Buchwald–Hartwig C–N coupling experiments of Ahneman et al. [25], which varied aryl halide, palladium ligand, base and additive. The label is the yield in %, from 0 to 100. One sample is a whole reaction, written as its reactants and product.

Buchwald–Hartwig yield. This is the version released with Schwaller et al. [26], from that paper's public repository: 3,955 reactions. Each is written as aryl halide, 4-methylaniline, ligand, base and additive going to a product, generated from the

aryl halide by a C–N coupling template that succeeds on all 3,955. They fall into 22 additive groups of 178 to 180 reactions.

Buchwald–Hartwig plates. This is the data table of the public repository of Ahneman et al. [25]: 4,599 rows over three plates, with product SMILES from the file. The 287 rows without an aryl halide are dropped. The 180 remaining rows without an additive form their own group, giving 24 additive groups of 177 to 180 reactions. The 4,312 reactions kept include all 3,955 of the first set, with the same aryl halide, ligand, base, additive and yield, plus 357 more: the 180 without an additive and 177 with 5-phenyl-1,2,4-oxadiazole as the additive.

S.5.4 Antibodies

The Koenig and nanobody sets were downloaded as CSV files from the FLAb benchmark repository [27], the humanness set from the BioPhi publication repository and the trastuzumab set from its own repository. The FLAb and BioPhi files come from each repository's main branch, with no commit pinned. Each file has been changed only once. The Koenig and nanobody files were changed in FLAb commits `c41e56e` and `005e6c1`, both on 6 August 2025, and the two humanness files in BioPhi commit `c0797f2` on 24 August 2021. The local copies of the Koenig and humanness files are identical to those commits.

The nanobody and humanness sets are grouped by sequence identity, defined for sequences a and b as

$$\mathrm{id}(a, b) = 1 - \frac{\mathrm{lev}(a, b)}{\max(|a|, |b|)}, \tag{S.1}$$

where lev is the Levenshtein (edit) distance and $|a|$ is the length in residues. Following Kenlay et al. [28], sequences are clustered by single linkage at id ≥ 0.95. The 672 nanobodies fall into 309 clusters and the 564 humanness antibodies into 461.

Koenig expression. This deep mutational scan of the anti-VEGF antibody G6 [29], taken from FLAb, has 4,275 variants with both chains (heavy chains of 120 residues, light chains of 108). Each carries exactly one mutation, 2,261 in the heavy chain and 2,014 in the light chain, at 225 distinct positions. The label is the unitless expression enrichment ratio, from 0.00098 to 3.178.

Nanobody thermostability. This is the NbThermo set [30], taken from FLAb: melting temperatures T_m from 39.0 to 98.2 °C for 672 single-domain antibodies with no light chain. The 672 rows hold 556 distinct sequences; 116 appear twice, and 106 of those pairs carry different melting temperatures, up to 8.8 °C apart. These repeats are kept, so the labels carry disagreement between measurements before any noise is added.

Humanness. This is the list of 564 therapeutic antibodies from IMGT/mAb-DB released with BioPhi [31], using the first sequence listed for each. An antibody is labeled human (1) only when its species field is exactly "Human" (198 antibodies). The other 366 are labeled non-human (0). The 564 antibodies hold 506 distinct sequences, mostly because an antibody and its drug conjugate share a sequence. One sequence is shared by three antibodies with different labels, two humanized and one human. All three sit in one identity cluster, so they never fall on both sides of a split.

Trastuzumab–HER2 binding. This is the library of trastuzumab variants in the ten-residue CDR-H3 (the third complementarity-determining region of the heavy chain), screened for HER2 binding by Chinery et al. [32]. The released training, validation and test files are joined and re-split. All 524,346 variants have a 10-residue CDR-H3 and are kept; 32.8% are binders. From these, 5,000 variants were drawn once with seed 42, stratified so that each combination of label and number of mutated CDR-H3 residues keeps its share; the sample holds 1,639 binders (32.8%). Each 120-residue heavy chain is rebuilt by inserting the variant CDR-H3 into the trastuzumab framework, and paired with the fixed trastuzumab light chain.

S.5.5 Nucleotide Sequences

Both nucleotide tasks come from the revised downstream tasks of the Nucleotide Transformer [33] on Hugging Face (`InstaDeepAI/nucleotide_transformer_downstream_tasks_revised`). The download is not pinned to a revision; the files used are those of revision `851f994`, dated 30 June 2025, the latest at the time of writing. The released training and test files are merged, and folds are grouped by the chromosome read from each sequence's name. Nothing else is filtered. Both tasks cover chromosomes 1 to 21, X and Y; chromosome 22 is absent.

H3K4me3. Each sample is a 1,000-base sequence, labeled by whether it carries the H3K4me3 histone mark. The merged files hold 18,244 sequences, 9,122 per class. The dataset card lists 30,000 training sequences; the count here is that of the files.

TATA promoter. Each sample is a 300-base sequence, labeled by whether it is a promoter with a TATA box. The merged files hold 5,274 sequences, 2,637 per class.

S.5.6 Cancer Genomics (TCGA)

The two cancer datasets use TCGA data [34] from UCSC Xena [35]. Expression is each cohort's Illumina HiSeq RNA-seq matrix on the Xena TCGA hub (version 2017-10-13): the same 20,530 genes in every cohort, unfiltered, as $\log_2(x + 1)$ of the RSEM-normalized count [36]. Only primary tumors are kept, one sample per patient. Missing expression values are set to 0. No batch correction is applied, because every fold holds out whole tissue source sites and a correction fitted on the training sites cannot adjust an unseen site, and a correction fitted on all sites would already have seen the test samples.

Lung adenocarcinoma against lung squamous cell carcinoma. The adenocarcinoma [37] and squamous cell carcinoma [38] cohorts are stacked, with the cohort as the label. Dropping 110 normal-tissue and 2 recurrent-tumor samples leaves 515 adenocarcinomas and 502 squamous tumors, 1,017 in all, from 68 tissue source sites.

Breast cancer PAM50 subtype. The label is the PAM50 subtype call [39] from the Xena breast cancer clinical matrix [40]. Tumors called normal-like or without a call are dropped, leaving 821 of 1,097 primary tumors with expression data: 421 luminal A, 192 luminal B, 141 basal-like and 67 HER2-enriched, from 19 tissue source sites.

S.6 Data and code availability

The software behind this thesis includes NoiseInject, the label-noise package released with Chapter 3, and the KIRBy repository, which implements the adaptive hybrid representation and holds the experiments of Chapters 4, 5 and 6 and the assay-dataset runs of Chapter 3.

NoiseInject is available as an open-source Python package under an MIT license [41]. Its noise conditions and metrics are described in Section 2.5.1.

- Project name: NoiseInject
- Project home page: https://github.com/adpunt/noise_inject
- Archived version: https://doi.org/10.5281/zenodo.20532710
- Version: 1.0.0
- Operating system(s): Platform independent
- Programming language: Python 3.7 or later
- Other requirements: NumPy, pandas, SciPy and scikit-learn; see the repository's `requirements.txt`
- License: MIT
- Any restrictions to use by non-academics: None

The QM9 results of Chapter 3, with that chapter's hyperparameter tuning, variance decomposition and figure scripts, come from a separate repository in Python and Rust, described in Chapter 3 under "QM9 Data Pipeline".

- Project name: qsar_qm_models
- Project home page: https://github.com/adpunt/qsar_qm_models
- Archived version: not yet archived
- Operating system(s): Linux and macOS
- Programming language: Python and Rust (2021 edition)
- Other requirements: RDKit installed through conda, to which the Rust program links; PyTorch and PyTorch Geometric
- License: GNU GPL v3.0
- Any restrictions to use by non-academics: those of the GNU GPL v3.0

KIRBy is written in Python and works on feature matrices computed beforehand, so the hybrid search applies to representations from any domain. It holds the hybrid search, loaders for the datasets in Additional file S.5, generators for the representations, the similarity analyses and learning-with-noisy-labels methods of Chapter 6, and every experiment script. Label noise is added through NoiseInject, so the noise definitions match Chapter 3.

- Project name: KIRBy
- Project home page: `https://github.com/adpunt/KIRBy`
- Archived version: [Zenodo DOI]
- Operating system(s): Linux and macOS
- Programming language: Python 3.8 or later; several pretrained encoders need Python 3.10 or later
- Other requirements: NumPy, pandas, SciPy, scikit-learn, and RDKit installed through conda; each application area has an optional install
- License: [license]
- Any restrictions to use by non-academics: [depends on the license]

All datasets but one are public. QM9 [20] is deposited on Figshare [21]. The LogD and Caco-2 efflux data are from the OpenADMET–ExpansionRx release under CC-BY-4.0 [17]. The hERG K_i data come from ChEMBL [15], released under CC BY-SA 3.0, and can be regenerated with the protocol in Additional file S.5. The rat liver microsome clearance and MDR1–MDCK efflux data are the Biogen ADME set [18], on Polaris [19] under CC-BY-4.0. The reaction-yield, antibody, nucleotide and tumor gene-expression data come from the public releases cited in Additional file S.5. Pretrained model weights come from their original sources and are not redistributed.

The exception is Lhasa Limited's curated hERG training and test sets [42] (Additional file S.5), used for feature selection with data held at separate sites (Chapter 5). They are proprietary and not redistributed. The comparison with FLuID also cannot be run without Lhasa's FLuID code. Every other result in this thesis can be reproduced from public data and the code above.

Lhasa Limited supplied the curated hERG data and the FLuID code [42]. The use of generative AI is described in the declaration in the front matter.

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